Physics-Guided Probabilistic Glucose Forecasting in Type 1 Diabetes: An Integrity-Checked Multi-Horizon Evaluation with Stable, Dose-Consistent Patient-Specific Insulin Sensitivity

Samyak Jain, Dhriti Goswami, Yugananda Malhotra
Bharati Vidyapeeth's College of Engineering New Delhi, 110063, India
sammyyakk@gmail.com; goswami.dhritid@gmail.com; yugnanda_malhotra@yahoo.com

***Abstract*—Background and objective: Glucose dynamics in type 1 diabetes (T1D) are nonlinear, delayed, and patient-specific, influenced by meals and insulin. Clinically meaningful forecasting requires valid prediction horizons and comparison with persistence baselines. This study proposes a physics-guided probabilistic forecaster under an integrity-checked protocol integrating deep sequence learning with mechanistic physiological modeling for personalized, multi-horizon glucose forecasting.**

Methods: A gap-aware pipeline was developed using the OhioT1DM dataset (12 subjects), which admitted only windows whose targets were real CGM readings exactly 30, 60, 90, and 120 min ahead (26,498 test windows). The model couples a compact Transformer encoder to a differentiable Bergman minimal model, extended with subcutaneous insulin and gut absorption states, through a physics residual and a mechanistic prior, with patient-specific parameter estimation and a non-crossing quantile head for uncertainty. Performance was assessed under personalized (temporal) and leave-one-subject-out (LOSO) protocols using RMSE, skill over persistence, Clarke/Parkes error grids, hypoglycemia-alarm sensitivity, and feature attribution. A pre-specified ablation with subject-level Wilcoxon tests and Holm correction tested the contribution of physics, and pre-specified checks tested whether insulin sensitivity (S_I) was subject-specific, stable, and dose-consistent.

Results: Personalized RMSE was 18.84, 30.52, 38.60, and 44.46 mg/dL at 30, 60, 90, and 120 min (19.4–21.2% skill over persistence); at 30 min all 12 subjects improved, and under LOSO all 12 still beat persistence at 30 min by MAE (10 of 12 by RMSE). In a post hoc five-seed replication, physics reduced MAE at every horizon (30-min MAE 13.37 ± 0.11 vs. 13.88 ± 0.25 mg/dL across seeds; 2.38 mg/dL lower at 120 min; Holm $p = 0.002$), although the pre-specified single-seed test was not significant (13.25 vs. 13.51 mg/dL; Holm $p = 0.105$). It also yielded an insulin-sensitivity estimate (S_I) that varied between subjects (CV 35.8%), was stable within them (ICC 0.89), and fell with total daily insulin ($\rho = -0.84$), though not under LOSO. The 5-min glucose rate of change was the top predictor under both attribution methods. At 30 min, the median forecast detected 55.1% of hypoglycemic readings; a 10th-percentile alarm raised this to 91.8% at a precision of 0.347.

Conclusion: Physiological structure yielded a stable insulin-sensitivity estimate, consistent with insulin dose under personalized evaluation, and small but consistent accuracy gains. Calibrated predictive tails, not error-grid scores, determine hypoglycemia detection.

***Index Terms*—continuous glucose monitoring, type 1 diabetes, physics-guided machine learning, Bergman minimal model, probabilistic forecasting, hypoglycemia, insulin sensitivity**

I. Introduction

A. Background

Type 1 diabetes (T1D) is a disease involving the immune-mediated destruction of pancreatic β-cells. Individuals with T1D therefore require lifelong exogenous insulin replacement and glucose monitoring, and disease-specific education, diet, and lifestyle modification remain cornerstones of management alongside insulin therapy. Glucose–insulin dynamics, however, are governed by nonlinear, patient-specific physiology, as described mechanistically through the Bergman minimal model [1]. Despite advancements in continuous glucose monitoring (CGM) hardware, it remains very difficult to build a system that converts raw diabetes data into personalized, actionable medical recommendations [2].

This has driven the development of machine-learning approaches for CGM forecasting, progressing from recurrent networks to attention-based Transformer architectures [3-6], which can achieve strong short-horizon accuracy but offer clinicians little interpretability [7-8]. This limitation motivated a shift toward physiology-aware approaches that add a physics-consistency penalty derived from mechanistic models, following the Physics-Informed Neural Network (PINN) paradigm [9-12]. However, some of these models were validated only in simulation [10]. More recent systems introduced patient-specific digital twins [13-15] and prospective evaluation [16-17]. Yet translating predictions into safe, actionable recommendations remains challenging, with reinforcement-learning approaches still considered insufficiently clinically safe [18].

The field has therefore advanced along several parallel tracks (interpretable risk prediction, physiology-constrained forecasting, digital-twin construction and validation, and increasingly capable sequence-modelling backbones) that remain largely disconnected from one another.

B. Literature Review

Systematic reviews of Type 1 diabetes digital twins note that existing systems are not yet comprehensive, multi-scale virtual replicas, and most model glucose–insulin metabolism in isolation rather than the full patient [2].

Explainable short-horizon hypoglycemia/hyperglycemia prediction models added SHAP-based feature attributions to opaque risk classifiers, demonstrating interpretable patient-level prediction [7]. Other work used SHAP attributions to guide bounded, one-step behavioural recommendations on OhioT1DM, though the approach remained exploratory and was not considered clinically safe [18]. Explainable postprandial glucose modeling attributed meal-response forecasts to specific inputs, achieving personalized, transparent prediction, though remaining purely data-driven with no physiological constraint [19].

Physiology has also been embedded directly into the learning objective. Early PINN-based glucose predictors added a physics-consistency penalty to sequence learning, improving short-term in-silico prediction, though validation stayed simulator-only [10]. Hybrid physiology and deep-learning architectures embedded glucose–insulin dynamics into both architecture and training, achieving reliable predictions under limited real-subject data, but remained forecasting models rather than virtual patients [11]. Physics–LSTM hybrids addressed recurrent black-box models' inability to distinguish meal and insulin effects, though still without a full virtual-patient representation [12].

Integrating physiology-based models with LSTM learning produced patient-specific virtual subjects for in-silico therapy testing, though as a research framework rather than a clinically validated system [13]. An open-source digital-twin methodology identified a personalized glucose–insulin model directly from routine CGM, insulin, and meal data for counterfactual simulation, though use remained mainly retrospective [14]. Simulation-to-reality augmentation used simulated patients as synthetic data generators to offset limited T1D data [20], and physiologically constrained digital twins were validated against real exercise data from 394 cases [15]. Randomized clinical trials showed that a twin-enhanced decision-support system improved time-in-range and reduced hypoglycaemia [16], and that cloud-based digital twins re-optimizing automated insulin delivery (AID) parameters every two weeks overcame the typical time-in-range plateau, though optimizing therapy periodically rather than forecasting glucose continuously [17].

In parallel, attention-based Transformer forecasting on OhioT1DM outperformed classical machine-learning baselines (XGBoost, 1D-CNN) for short-horizon prediction but remained a plain sequence model with limited interpretability [3]. Other work added interpretable multi-horizon attention over temporal and static covariates [4]. A hybrid Transformer–LSTM architecture improved short- and medium-horizon accuracy, though horizon coverage stopped short of 120 minutes [5], while long-horizon hybrids reached 120 minutes with richer clinical inputs but remained purely data-driven [6]. Personalized federated learning allowed models to be trained across multiple sites without sharing patient data, but did not consider on-device efficiency or interpretability [21]. Lightweight Transformers made edge deployment more practical, although they lacked built-in uncertainty estimation and clear interpretability [22]. A Temporal Fusion Transformer improved RMSE by $\geq$13% over earlier baselines and provided feature and temporal importance with calibrated uncertainty, but its attention patterns were not validated against physiological ground truth [23]. Self-supervised Transformer pretraining learned reusable glucose representations for different downstream tasks [24]. Retrieval-augmented forecasting used similar historical episodes rather than raw sequences alone, but still lacked mechanistic grounding, SHAP-based explanations, and prospective clinical validation [25].

Three questions therefore remain open. First, how much of the accuracy reported on OhioT1DM survives an evaluation with verified horizons, a persistence baseline, and subjects rather than overlapping windows as the unit of analysis? Second, does embedding physiology improve point accuracy, or does its value lie in the parameters it produces, and are those parameters valid? Third, can a calibrated predictive distribution detect hypoglycemia that an accurate median forecast misses?

Table I situates this study among OhioT1DM forecasting approaches, grouped into non-physics learned or statistical sequence models and physics-informed hybrids; the proposed method belongs to the latter group.

C. Objectives and Contributions

- To develop a hybrid physics-guided forecaster that couples a Transformer with Bergman minimal-model constraints, delivering 30, 60, 90, and 120-minute glucose forecasts that consistently outperform a validated persistence baseline.
- To evaluate the framework under an integrity-checked protocol, using only real CGM targets and subjects as the unit of analysis, and to test it under both personalized and subject-disjoint (leave-one-subject-out) protocols on identical windows.
- To separate the value of physics from accuracy gain through a pre-specified ablation and a 5-seed replication, rather than reporting accuracy alone.
- To estimate patient-specific insulin sensitivity from routine CGM, insulin, and meal data, and to verify through pre-specified checks that it is individual, stable, and consistent with total daily insulin dose.
- To quantify forecast uncertainty and recover hypoglycemic events missed by the median forecast through calibrated lower-quantile alarms.

The paper is structured as follows. Section I introduces the background, related literature, and objectives. Section II details the methodology, including data acquisition and model development. Section III presents the experimental results and model evaluation. Section IV discusses the findings, Section V covers limitations and future work, and Section VI concludes. The appendix lists the patient-parameter vector; full notation tables are in the Supplementary Material.

II. Methodology

A. Integrity Requirements

Glucose-forecasting pipelines on OhioT1DM are exposed to several integrity pitfalls that can inflate or invalidate reported

TABLE I
LANDSCAPE COMPARISON OF OHIOT1DM BLOOD-GLUCOSE FORECASTING STUDIES BY ARCHITECTURE FAMILY (30/60/90/120-MINUTE HORIZONS)

Reference	Methodology	Cohort	RMSE@30	MAE@30	RMSE@60	MAE@60	RMSE@90	MAE@90	RMSE@120	MAE@120
Non-physics learned/statistical sequence models (6 subjects, official split)										
[75]	CNN/LSTM	2020	17.45	11.22	33.67	23.25	not reported†		not reported†	
[38]	Neural physiological encoder + LSTM	2018	17.80	–	–	–	not reported†		not reported†	
[74]	N-BEATS + BiLSTM	2020	18.22	12.83	31.66	23.60	not reported†		not reported†	
[73]	LSTM	2020	18.23	14.37	31.10	25.75	not reported†		not reported†	
[72]	NN-EIM	2020	18.63	10.08	32.27	17.69	not reported†		not reported†	
[26]	Stacked regression + activity	2020	18.99	13.73	33.39	25.04	not reported†		not reported†	
[67]	MS-LSTM	2020	19.05	13.50	32.03	23.83	not reported†		not reported†	
[27]	Multi-lag stacking	2020	19.21	13.93	33.65	25.31	not reported†		not reported†	
[35]	Dilated CNN (GluNet)	2018	19.28	–	31.83	–	not reported†		not reported†	
[28]	Latent-variable statistical	2020	19.37	13.76	32.59	24.64	not reported†		not reported†	
[29]	Multi-class LSTM (+2018 data)	2020	19.40	13.90	33.40	25.00	not reported†		not reported†	
[36]	SVR (multivariate)	2018	19.59	13.49	31.54	22.86	not reported†		not reported†	
[30]	Multitask CRNN	2020	19.79	13.62	33.73	24.54	not reported†		not reported†	
[37]	Gaussian process	2018	20.01	–	31.97	–	not reported†		46.24	–
[31]	Online ARMA + residual net	2020	20.03	14.52	34.89	24.61	not reported†		not reported†	
[32]	Personalised interpretable LSTM	2020	20.20	14.74	34.19	25.98	not reported†		not reported†	
[33]	Seq2Seq BiLSTM	2020	21.80	15.00	35.00	25.00	not reported†		not reported†	
Physics-informed + other-architecture hybrid (6 subjects, official split)†										
[34]	Physiological model + ANN	2018	19.33	–	31.72	–	not reported†		not reported†	
–	**Current Work: Hybrid physics-guided Transformer (Bergman penalty + prior)**	2020	**18.74**	**13.24**	**31.16**	**22.55**	**39.52**	**29.14**	**45.39**	**34.03**
–	**Current Work (same model)**	2018	**18.93**	**12.92**	**29.89**	**21.42**	**37.69**	**27.70**	**43.53**	**32.44**
–	*Persistence baseline (ours)*	2020	24.22	17.45	40.11	29.51	51.15	38.34	58.98	44.83
–	*Persistence baseline (ours)*	2018	22.52	16.28	36.19	26.92	46.22	34.76	53.89	40.97

† Cohort: OhioT1DM test subjects on the official split (2018: 559, 563, 570, 575, 588, 591; 2020: 540, 544, 552, 567, 584, 596); results are comparable only within a cohort. Some studies score forecasts against interpolated or extrapolated test values; the current work uses only recorded CGM readings. "Not reported"/–: not published for that horizon or metric (the BGLP challenges required only 30- and 60-min forecasts). All-subject and LOSO results: Tables IV and XIII.

accuracy: windows that span recording gaps, so that the nominal horizon differs from the true one; forward-filled or interpolated CGM values used as prediction targets; random splits of strongly overlapping windows, which place near-duplicate samples in both training and test sets; overlapping windows, rather than subjects, used as the unit of statistical analysis; and no persistence baseline, without which absolute error values cannot be interpreted. Published no-change baselines for OhioT1DM [39-40] make the last pitfall straightforward to check.

The pipeline used here was built to exclude each pitfall by design: each became a design constraint or one of 234 automated tests. Evaluation begins with persistence, subjects rather than windows are the unit of analysis, and every table is regenerated from stored predictions.

B. Dataset and Temporal Integrity

OhioT1DM contains 12 subjects in the 2018 and 2020 cohorts and 166,443 CGM observations sampled nominally every five minutes, together with basal and bolus insulin, carbohydrate, activity, sleep, and wearable channels [41-42]. The official split follows the same subjects forward in time and therefore measures personalized forecasting, not cross-subject generalization. The 2020 cohort's first test hour is excluded according to its challenge protocol; the 2018 cohort is not modified in this way.

All streams are aligned to an exact five-minute grid without dropping rows. Missing CGM runs of at most ten minutes may be interpolated for input construction, but interpolated values are never accepted as targets. A sequence is emitted only if its 24-step (two-hour) input satisfies the coverage rule and every target is a real observation exactly $h \in \{30, 60, 90, 120\}$ minutes after the anchor. The scaler is fit only on training-fold slots, with each physical slot counted once rather than once per overlapping window.

From 187,780 candidate windows, 141,100 (75.1%) survive: 36,395 are rejected for incomplete inputs and 10,285 for unavailable real-valued targets. Both evaluation protocols use the same 26,498 test windows. Consecutive windows share 23 of their 24 input steps, so the raw window count overstates the information available. Each window, together with its 120-min target, spans 48 five-minute steps (24 input plus 24 forecast), so windows share observed data unless they are at least 48 steps apart. Dividing the approximately 135,000 retained five-minute slots across the whole cohort (the 141,100 retained windows overlap and share these slots) by this block length gives

$$n_{\text{eff}} \approx \frac{135{,}000 \text{ slots}}{48 \text{ steps}} \approx 2{,}800, \tag{1}$$

a coarse approximation. An autocorrelation-based estimate [43], using the mean integrated autocorrelation time of 53.6 five-minute steps across subjects, gives $n_{\text{eff}} \approx 2{,}500$–$3{,}100$, consistent with Eq. (1). Both figures describe the whole cohort. The test set alone, on which all reported metrics and the hypoglycemia analysis rest, corresponds to about $26{,}498/48 \approx 552$ effectively independent windows, or about 494 using the 53.6-step autocorrelation time.

C. System Overview

Fig. 1 outlines the full pipeline from raw recordings to reported outcomes. Data collected from all twelve OhioT1DM subjects, comprising continuous glucose readings, insulin delivery, and meal logs, are first aligned to a uniform five-minute time grid; genuine gaps in recording are preserved, and only short gaps of at most ten minutes may be interpolated for model inputs, never for forecast targets. From this aligned record, two parallel processes follow. The grid-aligned insulin and meal signals are passed through a simplified physiological model of glucose regulation, based on the Bergman minimal model together with subcutaneous insulin and gut absorption compartments, which yields time-varying estimates of insulin on board, carbohydrate on board, remote insulin action, and the rate of glucose appearance. These physiological estimates are then combined with the glucose signal itself, insulin and meal timing, and time-of-day information to form a single 35-feature representation at each time step.

This feature representation is grouped into two-hour input windows, and windows are retained only when both the input span and the corresponding forecast target are fully intact, so that no window silently spans a genuine data gap. The resulting windows are evaluated under two complementary protocols: an official, personalized setting in which each subject's earlier data are available for training, and a twelve-fold leave-one-subject-out setting in which the model is evaluated on subjects it has never encountered. Each window is then processed by a compact Transformer encoder, which produces two outputs in parallel: an estimate of the subject's individual physiological parameters, and a probabilistic glucose forecast expressed as a median trajectory bounded by non-crossing lower and upper quantiles.

These two outputs are not independent. The estimated physiological parameters are used to re-simulate the same mechanistic equations introduced earlier, and the resulting trajectory is compared against the network's own forecast at multiple points across the two-hour horizon; the discrepancy between the two constitutes a physics-consistency signal that is folded back into training, alongside a regularization term that keeps the estimated parameters within physiologically plausible bounds. In this way, the mechanistic model contributes to the pipeline twice: once as an input feature, and once as a differentiable constraint on the learned forecast. Because physics enters training both as a residual penalty and as a gated prior, we refer to the model as a *hybrid physics-guided* forecaster (penalty + prior); "PINN" is used only for the literature paradigm. Finally, forecasts from both evaluation protocols are scored on a per-subject basis and summarized across the cohort, using both conventional accuracy metrics and clinically oriented measures of hypoglycemia detection.

D. Mechanistic Patient Model

The physiological core extends the Bergman minimal model and its insulin-sensitivity index [1], [44], as given in Eqs. (2)–(4):

$$\dot{G} = -(p_1 + X)\, G + p_1 G_b + \frac{R_a(t)}{V_G}, \tag{2}$$

$$\dot{X} = -p_2 X + p_3 (I - I_b), \tag{3}$$

$$\dot{I} = -nI + \frac{1000}{V_I} \cdot \frac{S_2}{t_{\max,I}}. \tag{4}$$

Here G is glucose, I plasma insulin, X remote insulin action, and subscript b denotes basal state. All rates are per minute. Basal subtraction in (3) is essential: at $I = I_b$, $X^* = 0$ and $G = G_b$ is an equilibrium. Driving the equation with total insulin instead would create a spurious downward glucose attractor.

The patient-specific insulin sensitivity is the steady-state gain, as given in Eq. (5):

$$S_I = \frac{p_3}{p_2}, \qquad X_{ss} = S_I (I - I_b), \tag{5}$$

with units $\mathrm{mL}\,\mu\mathrm{U}^{-1}\mathrm{min}^{-1}$. Parameters are mapped by sigmoid functions into physiologically sourced intervals informed by minimal-model studies in insulin-dependent diabetes [45] and regularized toward population anchors. Consequently, plausible range alone is not evidence of identifiability; external and stability checks are required.

Subcutaneous insulin absorption follows a two-compartment model [46], as given in Eqs. (6)–(8):

$$\dot{S}_1 = u_{\text{ins}}(t) - \frac{S_1}{t_{\max,I}}, \tag{6}$$

$$\dot{S}_2 = \frac{S_1}{t_{\max,I}} - \frac{S_2}{t_{\max,I}}. \tag{7}$$

For an impulse dose D, insulin on board is

$$S_1 + S_2 = D\, e^{-t/t_{\max,I}} \left(1 + \frac{t}{t_{\max,I}}\right), \tag{8}$$

which starts at D and decreases monotonically. Carbohydrate absorption follows compartmental meal models used in T1D simulation [46-48] and is represented by stomach and gut states, as given in Eqs. (9)–(11):

$$\dot{Q}_{\text{sto}} = -k_{\text{gri}} Q_{\text{sto}} + u_{\text{carb}}(t), \tag{9}$$

$$\dot{Q}_{\text{gut}} = k_{\text{gri}} Q_{\text{sto}} - k_{\text{abs}} Q_{\text{gut}}, \tag{10}$$

$$R_a(t) = f\, k_{\text{abs}} Q_{\text{gut}}(t), \tag{11}$$

so carbohydrate on board is $Q_{\text{sto}} + Q_{\text{gut}}$ and absorbed mass is conserved. For implementation, the non-glucose compartments are collected, as given in Eq. (12):

$$\mathbf{z} = [S_1, S_2, I, Q_{\text{sto}}, Q_{\text{gut}}, X]^{\mathsf{T}}, \quad \mathbf{u} = [u_{\text{ins}}, u_{\text{carb}}, I_b]^{\mathsf{T}}. \tag{12}$$

With parameters held constant over one grid interval, $\dot{\mathbf{z}} = A\mathbf{z} + B\mathbf{u}$ has the exact zero-order-hold update (a standard control-theory discretization technique, not attributed to a specific glucose-modeling source), as given in Eq. (13):

$$\mathbf{z}_{k+1} = A_d \mathbf{z}_k + B_d \mathbf{u}_k, \qquad \begin{bmatrix} A_d & B_d \\ 0 & I \end{bmatrix} = \exp\!\left(\Delta \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}\right). \tag{13}$$

This matrix-exponential formulation preserves the affine basal term through the explicit I_b input and avoids the accumulated integration error of a coarse explicit solver.

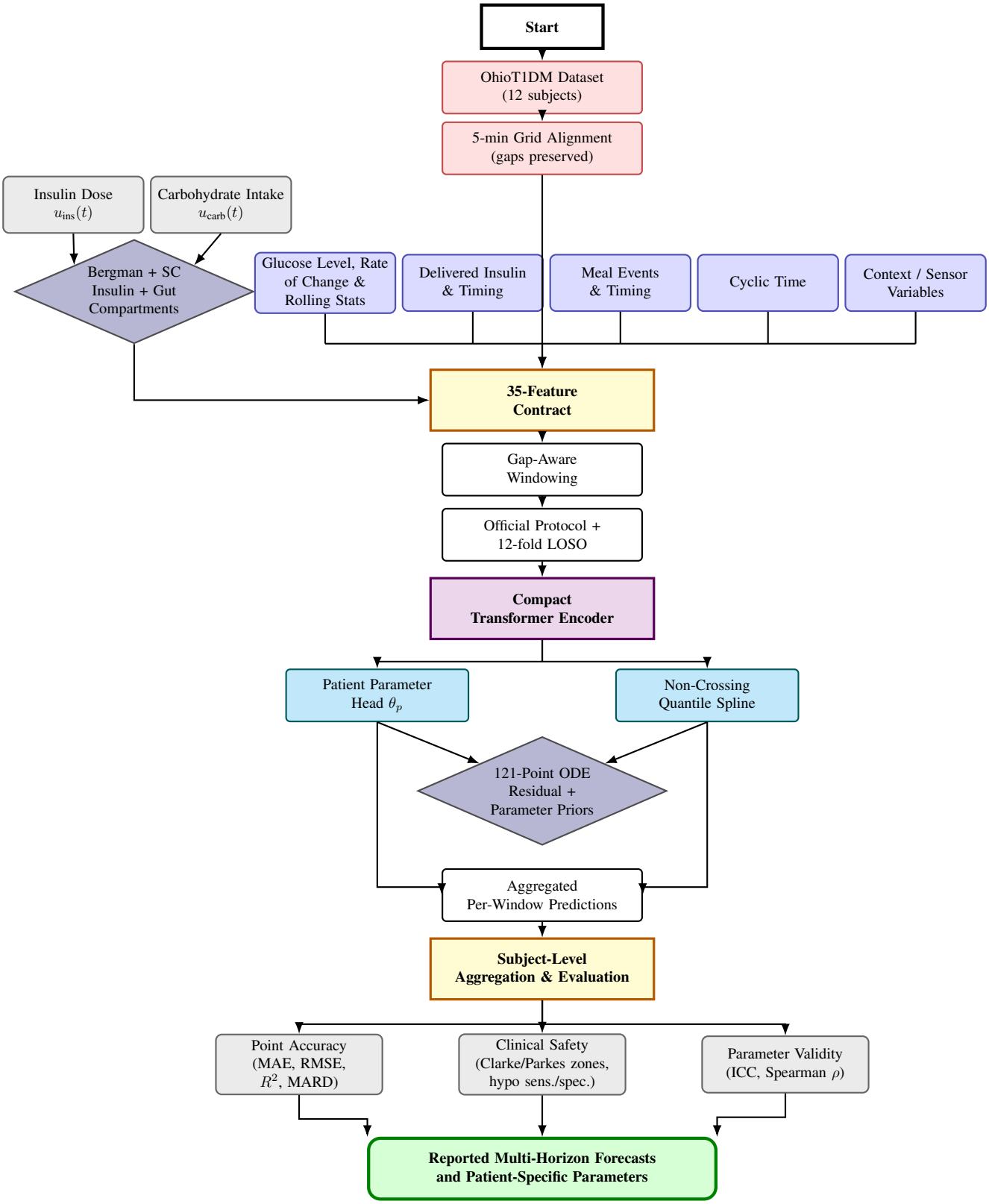


Fig. 1. *Integrity-checked forecasting pipeline of the hybrid physics-guided model. OhioT1DM glucose and related patient data are aligned to five-minute intervals and processed through two parallel paths: a deep-learning model and a physiological model. The deep-learning model predicts future glucose levels and estimates patient-specific parameters, while the physiological model represents insulin and glucose dynamics. Physiological information is used both as an input to the model and as a training constraint to improve prediction consistency.*

E. Feature Contract and Forecasting Network

Each timestep contains 35 named features in six groups: glucose level, multiscale rates of change and rolling statistics; delivered insulin and timing; meals and timing; mechanistic states (IOB, COB, plasma insulin, remote action, and glucose appearance); cyclic time; and context/sensor variables. A two-hour window $X \in \mathbb{R}^{24\times 35}$ is standardized using training-only statistics.

The forecasting network is deliberately small (approximately 79,000 parameters) because the effective sample size is limited. A linear projection and positional encoding feed a Transformer encoder [50]. Attention pooling, Eq. (14), gives

$$z = \sum_{t=1}^{24} \alpha_t h_t, \qquad \alpha_t = \frac{\exp(w^\top h_t)}{\sum_j \exp(w^\top h_j)}, \tag{14}$$

avoiding the equal weighting of old and recent measurements induced by mean pooling.

One head estimates the ten-component subject parameter vector θ_p. A second predicts a cubic B-spline trajectory, as given in Eq. (15):

$$G_\theta(t) = G_0 + \sum_{k=1}^{K} c_k\big(B_k(t) - B_k(0)\big), \tag{15}$$

which anchors every trajectory at the last observed glucose G_0. Median knot values are unconstrained, whereas lower and upper quantiles are generated by positive softplus offsets in glucose units; quantiles therefore cannot cross. A learned trust gate $g = \sigma(\gamma)$ controls the contribution of the mechanistic prior, with $g \approx 0.10$ at initialization. Predictions at all four horizons are evaluated from the spline.

F. Training Objective

The data objective combines Huber's robust loss [51] for the median with the pinball loss used in quantile regression [52] for lower and upper quantiles, as given in Eq. (16):

$$\mathcal{L}_{\text{data}} = \mathcal{L}_{\text{Huber}}(y, \hat{y}_{0.5}) + \sum_{q\in\{0.1,0.9\}} \rho_q(y - \hat{y}_q), \tag{16}$$

where $\rho_q(u) = u(q - \mathbb{1}_{u<0})$. Physics residuals are evaluated at 121 collocation points over the two-hour forecast and normalized by physiological scales, following the residual-minimization principle of PINNs [9]. The full objective is given in Eq. (17):

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}}\mathcal{L}_{\text{phys}} + \lambda_{\text{prior}}\mathcal{L}_{\text{prior}} + \lambda_{\text{tc}}\mathcal{L}_{\text{temporal}}. \tag{17}$$

Closed-form state updates replace a memory-intensive differentiable ODE solver. Adaptive loss weighting was motivated by work on gradient imbalance and uncertainty-based multi-objective learning [52-53]. The A7 configuration, selected on validation data rather than the test set, applies physics from epoch one; a data-first curriculum was rejected because validation optima repeatedly occurred before the curriculum became selectable.

G. Evaluation and Statistical Analysis

The official temporal protocol uses each subject's earlier data for training. LOSO trains on 11 subjects and evaluates the held-out subject on the identical test windows. Metrics are computed per subject and summarized as mean $\pm$ SD. Persistence predicts $\hat{G}(t+h) = G(t)$. Skill is $1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{persistence}}$, computed from the cohort-mean RMSEs.

Clinical evaluation includes Clarke and Parkes zones [54-56], consensus CGM ranges [58], coefficient-of-variation ratio, and Kovatchev low/high blood-glucose risk indices [58-59], together with hypoglycemia sensitivity, specificity, precision, and quantile calibration. Paired model comparisons use the Wilcoxon signed-rank test [61] across 12 subjects with Holm multiplicity correction [62]; bootstrap intervals resample subjects, never overlapping windows. Test–retest reliability of S_I is measured with one-way ICC(1) [63], and concurrent validity uses Spearman rank correlation [64].

H. Pre-specified Outcomes and Falsification Criteria

The outcomes and criteria below were fixed on 25 July 2026, before any test-set evaluation. The primary outcome is 30-minute MAE under the official protocol for the physiology-focused branch. Secondary outcomes include all horizons under both protocols, persistence skill, Clarke, Parkes, and PRED-EGA distributions, the ablation matrix, quantile calibration, and the three S_I validation checks. Table II records the criteria that were fixed in advance; unmet criteria are reported as failures rather than redefined after inspection.

No asymmetric hypoglycemia/hyperglycemia point-loss term is permitted, because such a term would directly optimize the safety table later used as evidence. The test set is evaluated only after model selection, and the official protocol is never described as subject-disjoint generalization.

I. Reproducibility, Hardware, and Latency

Models were implemented in PyTorch 2.10 (CUDA 12.8) and trained with AdamW (learning rate 5×10^{-4}, weight decay 0.03, batch size 256, gradient clipping at L_2 norm 1.0) for at most 70 epochs, with early stopping after 12 epochs without improvement in 30-min validation MAE. The final quantile-head run stopped earlier, after 11 epochs, with its best validation MAE at epoch 3. The hyperparameter search covered encoder depth $\in \{2, 4\}$, $d_{\text{model}} \in \{64, 128\}$, attention heads $\in \{4, 8\}$, $d_{\text{ff}} \in \{128, 512\}$, dropout $\in \{0.1, 0.2\}$, pooling (attention, mean, or last), and physics-loss weighting (fixed $\lambda_{\text{phys}} \in [0.01, 1.0]$ vs. Kendall homoscedastic uncertainty weighting).

Experiments ran on a single Linux machine with an NVIDIA GeForce RTX 2050 Laptop GPU (4 GB) and an Intel Core CPU. Across the seven training runs reported here (ablation arms A0–A4 and A7, and the final quantile-head model), training stopped after 11–20 epochs (mean 16.3), at 10.7–23.8 s per epoch, or 2.7–6.4 min per run. Inference latency for one window (batch size 1, 100 forward passes) was 15.25 ± 4.07 ms on GPU (median 13.21 ms, 95th percentile 25.07 ms) and 6.33 ± 0.18 ms on CPU (median 6.31 ms),

well within the 5-minute CGM sampling interval. Code, data-processing pipelines, and configurations are available at https://github.com/sammyyakk/digital-twin.

J. Feature Attribution Method

Integrated gradients (IG) [65] uses the standardized training mean as its zero baseline, as given in Eq. (18):

$$\mathrm{IG}_i(x) = (x_i - x_i') \int_0^1 \frac{\partial f(x' + \alpha(x - x'))}{\partial x_i}\, d\alpha. \qquad (18)$$

Because IG shows an unresolved completeness violation, every interpretation is cross-checked with model-agnostic permutation importance, following the perturbation principle popularized for random forests [66].

III. Results

A. Baseline Validation and Data Accounting

Persistence RMSE on the 2018 cohort is 22.52 ± 2.56 mg/dL at 30 minutes and 36.19 ± 3.26 mg/dL at 60 minutes, within 0.02 mg/dL of both independently published no-change baselines at 30 minutes and within 0.41 mg/dL of the published 60-minute value[39-40]. This close agreement validates the complete evaluation pipeline, including data parsing, five-minute grid alignment, horizon construction, persistence-target generation, RMSE computation, and subject-level aggregation (Table III). The retained-window accounting further confirms that the reported metrics are computed on the intended 26,498 common test windows.

B. Official-Protocol Forecast Accuracy

Table IV and Fig. 2 show that the model improves on persistence at every horizon. All 12 subjects improve at 30 minutes; subject-level MAE skill ranges from 12.2% to 31.6%. The decline in R^2 with horizon is expected because future meals, boluses, and activity are absent from the input.

C. Comprehensive Baseline Comparison on Identical Windows

Table V compares eleven models on the same 26,498 test windows from all 12 subjects under the official protocol:

1) **Persistence:** zero-order hold, $\hat{G}_{t+h} = G_t$.
2) **Bergman ODE, population parameters:** forward integration from the basal steady state with literature parameters [1], [44], without neural correction.
3) **Bergman ODE, per-subject fit:** parameters fitted per subject by nonlinear least squares/MAP on historical calibration intervals.
4) **AR(6):** linear autoregression on the last six CGM values (30-min lookback).
5) **ARX(6):** AR(6) with added insulin and carbohydrate inputs.
6) **ARX(6) + IOB/COB:** ARX(6) with mechanistic insulin- and carbohydrate-on-board added.
7) **Ridge (35 features):** ridge regression on the full 35-feature input used by A0 and A7.
8) **LSTM:** recurrent model on the 26 non-mechanistic features.
9) **Transformer, $A0_{\text{no-mech}}$:** the hybrid model's backbone trained on the same 26 features.
10) **Transformer, A0:** the same backbone with all 35 features, including mechanistic insulin- and carbohydrate-on-board.
11) **Hybrid physics-guided, A7:** A0 plus the Bergman prior, learned residual correction, and parameter head.

Mechanistic models alone performed worst: with population parameters the ODE drifted to a 30-min MAE of 29.70 mg/dL (54.18 mg/dL at 120 min), and per-subject fitting improved this only to 25.80 mg/dL (48.43 mg/dL at 120 min). AR(6) outperformed both non-mechanistic neural baselines at every horizon (30-min MAE 14.39 vs. 15.41 mg/dL for the LSTM and 14.96 mg/dL for $A0_{\text{no-mech}}$), and adding insulin and meal inputs (ARX) lowered MAE to 13.97 mg/dL. Adding mechanistic features to the Transformer reduced 30-min MAE by 1.45 mg/dL (14.96 to 13.51 mg/dL). To separate the contribution of richer inputs from that of the architecture, ridge regression on all 35 features and ARX(6) with IOB/COB were fitted on the same windows; they reached 30-min MAEs of 13.56 and 13.51 mg/dL, matching A0. At 30 min, therefore, most of the gain over the non-mechanistic neural baselines comes from the richer, mechanistically derived inputs, and A7 was 0.26–0.31 mg/dL lower than these linear models. The gap widened with horizon: A7's MAE was 1.34, 2.34, and 3.09 mg/dL below ridge regression at 60, 90, and 120 min. Differences from these two linear baselines were not tested for significance. A7 had the lowest MAE at every horizon in this comparison (13.25 mg/dL at 30 min).

D. Cross-Subject Generalization

At 30 minutes, LOSO remains better than persistence in all 12 subjects by MAE (10 of 12 by RMSE), but is consistently worse than the personalized protocol. As Fig. 3 shows, the personalization gap grows with horizon, consistent with increasing dependence on individual insulin kinetics and routine.

E. Physics Ablations

In the pre-specified single-seed ablation, no physics configuration significantly outperformed the no-physics A0 baseline after Holm correction (Table VI; Fig. 4). A7 was nominally best and improved 9 of 12 subjects, but its corrected $p = 0.105$, so the pre-specified criterion was not met. Holm correction was applied across the five arm-versus-A0 comparisons (A1–A4 and A7) on 30-min MAE. A post hoc 5-seed replication of A0 and A7, not part of the pre-specified plan (Section III-E1) found that, on seed-averaged per-subject errors, A7 has lower MAE at every horizon, by 0.51 mg/dL at 30 min and 2.38 mg/dL at 120 min (Holm $p = 0.002$ at each horizon). Per-patient parameters remain accuracy-neutral: A3 and population-fixed A4 differ by 0.07–0.23 mg/dL at 30/60 min, widening to 0.76 mg/dL at 120 min. In contrast with an earlier physiology-based OhioT1DM study that primarily reported forecast accuracy [34], this ablation separates parameter estimation from predictive gain.

TABLE II
PRE-SPECIFIED SUCCESS AND FALSIFICATION CRITERIA

Claim	Criterion fixed before test evaluation	Interpretation	Ref.
Beats persistence	30-minute MAE below persistence in at least 10 of 12 subjects	Requires broadly distributed improvement, not a pooled gain driven by a few subjects.	[41-42], [40]
Field-competitive point accuracy	30-minute MAE $\leq$ 13.5 mg/dL	A field-relative target, explicitly not a clinical acceptability threshold.	[42], [54-55], [67]
Physics improves accuracy	Physics arm beats A0 at 30 minutes and survives paired Wilcoxon testing with Holm correction	If unmet, physics is not credited with point-accuracy improvement.	[60-61]
S_I is subject-specific	Between-subject CV $>$ 10%	Rejects a parameter head collapsed to one population value.	[1], [44]
S_I is stable	ICC(1) $>$ 0.5 on disjoint within-subject blocks	Rejects a window-level nuisance variable that merely absorbs prediction error.	[63], [68]
S_I is dose-consistent	Spearman $\rho < 0$ versus total daily dose, with a subject-bootstrap 95% CI excluding zero	Evaluated separately under personalized and LOSO protocols.	[64], [68-69]

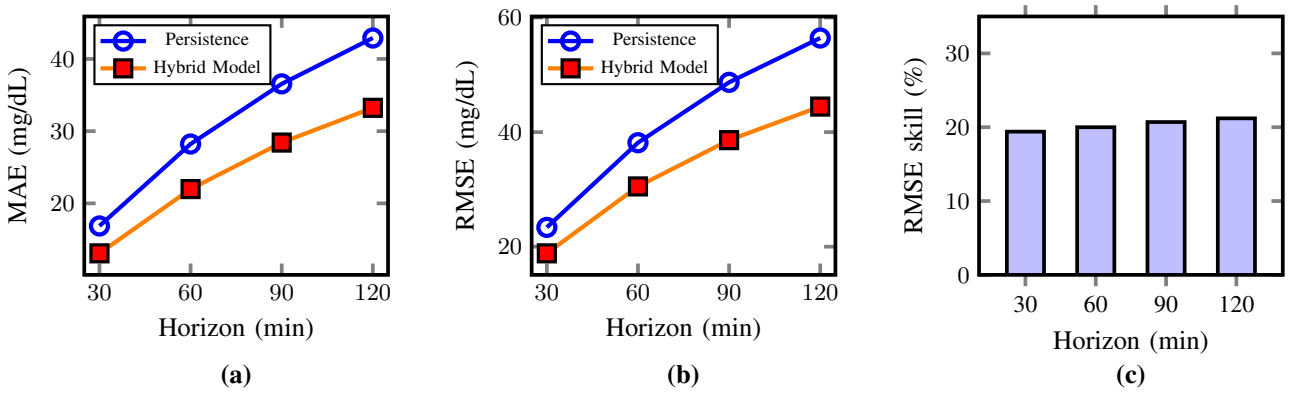


Fig. 2. *Official-protocol accuracy summary at 30/60/90/120 minutes. (a) and (b): MAE and RMSE for persistence versus the hybrid model both rise with horizon, but the gap between the two curves widens rather than closes, showing the model's advantage over naive persistence is not a short-horizon artifact. (c): the resulting RMSE skill stays essentially flat at 19–21% across all four horizons.*

TABLE III
WINDOW ACCOUNTING AFTER INTEGRITY FILTERING. ALL 12 SUBJECTS (2018 AND 2020 COHORTS); COUNTS ARE TOTALS ACROSS SUBJECTS, AND THE 26,498 TEST WINDOWS ARE SHARED BY BOTH EVALUATION PROTOCOLS.

Quantity	Value
Subjects / CGM observations	12 / 166,443
Candidate windows	187,780
Retained windows	141,100 (75.1%)
Incomplete input span	36,395
Missing real target	10,285
Common test windows	26,498
Effective independent windows, whole cohort	$\approx$ 2,800
Effective independent test windows	$\approx$ 552 ($\approx$ 494)

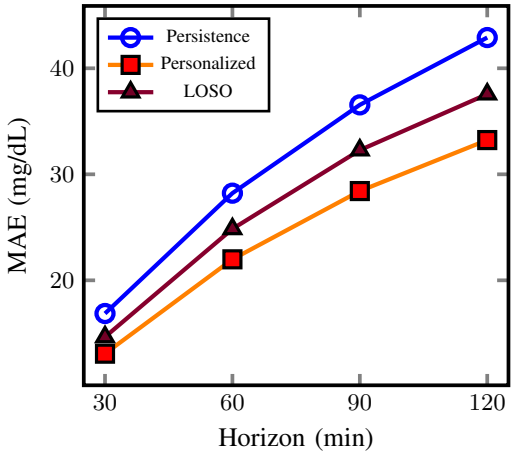


Fig. 3. *Matched-protocol comparison across 30, 60, 90, and 120-minute horizons. Persistence, personalized, and LOSO models are evaluated on identical windows. LOSO keeps a roughly constant MAE advantage of about 12% over persistence at every horizon, while its gap to the personalized model widens from 1.6 to 4.3 mg/dL, indicating a greater benefit from within-subject history for extended forecasting.*

Beyond this modest accuracy gain, the value of physics lies in access to a stable physiological parameter and informative mechanistic states.

Arms A5 and A6. Two planned arms were dropped for technical reasons:

- **A5 (simulation pretraining)** would have pretrained on UVA/Padova simulator traces, but the legacy meal generator re-sampled meal times inside the integration loop, producing about 0.6 instead of 3 meals per day; A5 was blocked to avoid training on invalid synthetic data.
- **A6 (legacy heuristic kernels)** would have compared Bergman-derived IOB/COB with legacy kernels that had an unnormalized carbohydrate decay and a time-reversed insulin curve. Instead, the value of the compartmental states is quantified by feature attribution (Section III-I), where they carry 27.1% of total attribution.

TABLE IV

OFFICIAL PERSONALIZED PROTOCOL, ALL 12 SUBJECTS, FINAL MODEL (MEDIAN OF THE QUANTILE-HEAD RUN): MEAN ± SD ACROSS SUBJECTS (MG/DL UNLESS NOTED). THESE VALUES COME FROM THE FINAL-MODEL RUN, WHICH WAS TRAINED SEPARATELY FROM ABLATION ARM A7 IN TABLES V AND VI; THIS EXPLAINS THE SMALL DIFFERENCES (E.G., 30-MIN MAE 13.08 HERE VS. 13.25 THERE).

Horizon	Persistence MAE	Model MAE	Persistence RMSE	Model RMSE	RMSE skill	R^2	MARD
30 min	16.87 ± 1.92	**13.08 ± 1.73**	23.37	**18.84 ± 2.58**	19.4%	0.885	8.79%
60 min	28.22 ± 3.24	**21.98 ± 2.96**	38.15	**30.52 ± 3.98**	20.0%	0.700	14.76%
90 min	36.55 ± 4.00	**28.42 ± 3.89**	48.69	**38.60 ± 5.05**	20.7%	0.523	19.14%
120 min	42.90 ± 4.46	**33.23 ± 4.45**	56.43	**44.46 ± 5.62**	21.2%	0.368	22.37%

TABLE V

COMPREHENSIVE BASELINE COMPARISON ACROSS 12 SUBJECTS EVALUATED ON IDENTICAL TEST WINDOWS ($n = 26{,}498$). METRICS REPORTED AS MEAN ± SD ACROSS SUBJECTS (MG/DL). THE A7 ROW IS THE SINGLE-SEED ABLATION RUN (AS IN TABLE VI), NOT THE FINAL MODEL IN TABLE IV.

	30 min		60 min		90 min		120 min	
Model	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Persistence	16.87 ± 1.92	23.37 ± 2.85	28.22 ± 3.24	38.15 ± 4.58	36.55 ± 4.00	48.69 ± 5.58	42.90 ± 4.46	56.43 ± 6.21
Bergman ODE (population)	29.70 ± 9.67	39.67 ± 12.67	43.71 ± 12.84	57.08 ± 16.10	50.72 ± 13.66	65.26 ± 16.79	54.18 ± 13.78	69.20 ± 16.77
Bergman ODE (per-subject NLS)	25.80 ± 8.36	34.72 ± 10.45	38.09 ± 9.56	50.41 ± 11.78	44.70 ± 8.99	58.47 ± 11.01	48.43 ± 8.35	63.03 ± 10.29
AR(6)	14.39 ± 1.52	20.11 ± 2.65	25.52 ± 2.90	33.88 ± 3.90	33.18 ± 3.66	42.92 ± 4.62	38.30 ± 3.90	48.64 ± 4.92
ARX(6)	13.97 ± 1.51	19.61 ± 2.76	24.48 ± 2.86	32.67 ± 3.98	32.24 ± 3.47	41.82 ± 4.47	37.68 ± 3.69	48.00 ± 4.77
ARX(6) + IOB/COB	13.51 ± 1.50	19.09 ± 2.69	23.71 ± 2.86	31.86 ± 3.77	31.39 ± 3.51	41.04 ± 4.25	36.84 ± 3.74	47.22 ± 4.50
Ridge (35 features)	13.56 ± 1.55	19.10 ± 2.60	23.63 ± 3.07	31.80 ± 3.81	31.11 ± 3.89	40.94 ± 4.65	36.47 ± 4.18	47.04 ± 5.17
LSTM (non-mechanistic)	15.41 ± 2.16	21.46 ± 2.83	26.08 ± 3.87	35.24 ± 4.56	33.59 ± 4.77	44.97 ± 5.76	38.81 ± 5.30	51.54 ± 6.63
Transformer ($A0_{\text{no-mech}}$)	14.96 ± 1.82	21.08 ± 2.55	26.06 ± 3.15	35.56 ± 4.15	34.46 ± 3.81	46.08 ± 5.11	40.06 ± 4.09	53.03 ± 5.63
Transformer (all features, A0)	13.51 ± 1.63	19.38 ± 2.41	22.82 ± 2.88	31.61 ± 3.71	29.74 ± 3.74	40.32 ± 4.71	34.93 ± 4.20	46.70 ± 5.24
Hybrid physics-guided (A7)	**13.25 ± 1.69**	**19.06 ± 2.51**	**22.29 ± 2.93**	**30.89 ± 3.89**	**28.77 ± 3.75**	**39.04 ± 4.83**	**33.38 ± 4.21**	**44.78 ± 5.46**

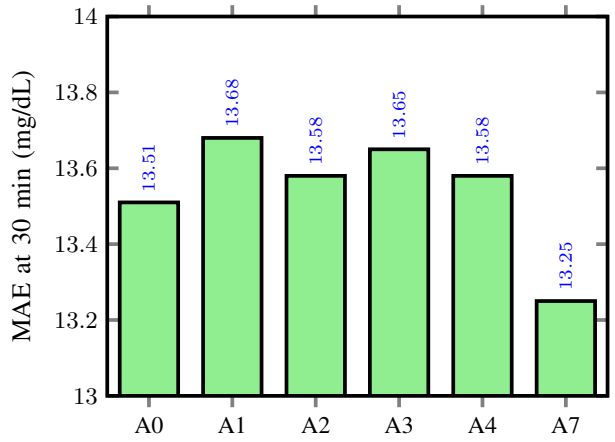


Fig. 4. *Point-accuracy ablation at 30 minutes comparing six physics configurations (A0-A4,A7), from no physics (A0) to full physics from epoch 1 (A7). A7 has the lowest MAE (13.25 mg/dL) versus 13.51 mg/dL for A0 in the primary run (seed 42). Multi-seed replication across 5 random seeds confirms this advantage is statistically significant across all horizons under paired Wilcoxon testing (Table VII).*

1) Multi-Seed Replication and Statistical Significance: As a post hoc check of seed sensitivity, A0 and A7 were each retrained from scratch with five seeds $\{42, 101, 202, 303, 404\}$ under the official protocol. We report the mean and SD of MAE and RMSE across seeds, and paired two-sided Wilcoxon signed-rank tests on seed-averaged per-subject errors with Holm–Bonferroni correction across the four horizons, applied separately to MAE and RMSE (Table VII).

A7 significantly outperformed A0 at every horizon ($p_{\text{Holm}} = 0.00195$ for both MAE and RMSE), with an MAE advantage of 0.51 mg/dL at 30 min and 2.38 mg/dL at 120 min. Physics also reduced variability across seeds: the seed SD was 0.11 vs. 0.25 mg/dL at 30 min and 0.64 vs. 1.07 mg/dL at 120 min for A7 vs. A0.

F. Clinical Accuracy and Hypoglycemia Detection

Clinical agreement remains high across horizons: Clarke A+B decreases from 99.0% at 30 minutes to 94.3% at 120 minutes (Fig. 5). The Parkes error grid, pooled across test windows, shows the same pattern: zone A+B falls from 99.8% at 30 minutes to 95.9% at 120 minutes, and zone E is 0% at every horizon.

Because Clarke and Parkes grids were designed for point-in-time sensor measurements rather than forecasts, we also applied Prediction Error Grid Analysis (PRED-EGA) [71], which classifies predictions as Accurate, Benign, or Erroneous separately in hypoglycemia ($\leq$ 70 mg/dL, following the PRED-EGA definition), euglycemia (70–180 mg/dL), and hyperglycemia ($>$ 180 mg/dL) (Table VIII). Clinically acceptable predictions (Accurate + Benign) were at least 99.4% at every horizon, and Erroneous predictions were at most 0.6%, reaching that value at 120 min. In hyperglycemia, 96.2% of predictions were Accurate at 30 min and 86.1% at 60 min; in hypoglycemia, Erroneous predictions were 0.00% at 30 min and 0.13% at 60 min.

However, as observed with Clarke zone A, standard point-forecast metrics reward predictions close to the center and can underestimate acute excursions. Thus, for the final model at 30 minutes, 89.8% of predictions in Clarke zone A (mean of per-subject values; Fig. 5) and 99.96% PRED-EGA clinically acceptable can still coexist with the median point forecast missing 45% of hypoglycemic readings (pointwise sensitivity 0.551; Table IX). Within the hypoglycemic range itself, only

TABLE VI

ABLATION MATRIX: OFFICIAL PROTOCOL, ALL 12 SUBJECTS, ONE TRAINING RUN PER ARM (SEED 42). MAE AT ALL HORIZONS; CLARKE A AND HYPOGLYCEMIA METRICS AT 30 MIN; HYPO BIAS IS THE MEAN ERROR FOR GLUCOSE BELOW 70 MG/DL. A3 AND A4 DIFFER ONLY IN PER-PATIENT PARAMETERIZATION; PHYSICS WEIGHTING AND CURRICULUM ARE HELD FIXED. HYPOGLYCEMIA VALUES DIFFER FROM TABLE IX, WHOSE A7 ROW IS THE FINAL MODEL OF TABLE IV AND WHOSE A0 ROW WAS RETRAINED WITH THE SAME QUANTILE HEAD (SECTION III-G2).

Arm	Configuration	MAE30	MAE60	MAE90	MAE120	Clarke A	Hypo sens.	Hypo bias	Gives S_I
A0	Transformer, no physics	13.51	22.82	29.74	34.93	89.4%	0.637	+7.29	no
A1	Physics penalty, fixed $\lambda = 0.1$	13.68	23.47	30.93	36.22	88.9%	0.502	+11.29	no
A2	Adaptive penalty, per patient	13.58	23.35	30.59	35.74	89.0%	0.499	+11.03	yes
A3	Hybrid prior, curriculum	13.65	23.12	30.02	34.83	89.0%	0.485	+10.95	yes
A4	Hybrid prior, curriculum, population parameters	13.58	23.35	30.57	35.59	89.1%	0.517	+10.18	no
A7	Hybrid, full physics from epoch 1	**13.25**	**22.29**	**28.77**	**33.38**	**89.8%**	0.608	+8.61	yes

TABLE VII

POST HOC MULTI-SEED REPLICATION (5 SEEDS: 42, 101, 202, 303, 404) COMPARING A0 (NO PHYSICS) AND A7 (HYBRID PHYSICS-GUIDED), WITH PAIRED WILCOXON SIGNED-RANK TESTS ON SEED-AVERAGED PER-SUBJECT ERRORS. VALUES ARE MEANS OVER THE FIVE SEEDS AND THEREFORE DIFFER FROM THE SINGLE-SEED (SEED 42) VALUES IN TABLES V AND VI.

	A0 (Transformer, no physics)		A7 (Hybrid physics-guided)		Paired Wilcoxon Test (Seed-Averaged)			
Horizon	MAE (mean ± SD)	RMSE (mean ± SD)	MAE (mean ± SD)	RMSE (mean ± SD)	MAE Diff	p_{raw}	p_{Holm}	Significant?
30 min	13.88 ± 0.25	19.72 ± 0.25	$\mathbf{13.37 \pm 0.11}$	$\mathbf{19.17 \pm 0.16}$	-0.51	0.00049	0.00195	**Yes** ($p < 0.01$)
60 min	23.78 ± 0.64	32.70 ± 0.71	$\mathbf{22.60 \pm 0.32}$	$\mathbf{31.23 \pm 0.44}$	-1.18	0.00049	0.00195	**Yes** ($p < 0.01$)
90 min	31.21 ± 1.00	42.08 ± 1.17	$\mathbf{29.28 \pm 0.53}$	$\mathbf{39.61 \pm 0.70}$	-1.93	0.00049	0.00195	**Yes** ($p < 0.01$)
120 min	36.46 ± 1.07	48.46 ± 1.22	$\mathbf{34.08 \pm 0.64}$	$\mathbf{45.47 \pm 0.86}$	-2.38	0.00049	0.00195	**Yes** ($p < 0.01$)

TABLE VIII

PRED-EGA (SIVANANTHAN ET AL. [71]) CLINICAL PREDICTION ERROR GRID RESULTS ACROSS 12 SUBJECTS (MEAN ± SD ACROSS SUBJECTS)

Metric	30 min	60 min	90 min	120 min
Overall Accurate (%)	90.8 ± 3.3	77.8 ± 7.0	68.6 ± 8.1	61.9 ± 8.3
Overall Benign (%)	9.2 ± 3.3	21.9 ± 7.0	30.9 ± 8.1	37.5 ± 8.2
Overall Erroneous (%)	0.04 ± 0.07	0.28 ± 0.28	0.50 ± 0.39	0.60 ± 0.44
Clinically Acceptable (%)	$\mathbf{99.96 \pm 0.07}$	$\mathbf{99.72 \pm 0.28}$	$\mathbf{99.50 \pm 0.39}$	$\mathbf{99.40 \pm 0.44}$
Hypoglycemia Accurate (%)	65.9 ± 20.1	33.8 ± 16.5	18.3 ± 12.0	10.9 ± 8.2
Hypoglycemia Erroneous (%)	0.00 ± 0.00	0.13 ± 0.44	0.24 ± 0.54	1.59 ± 3.82
Hyperglycemia Accurate (%)	96.2 ± 2.8	86.1 ± 6.8	75.6 ± 9.6	66.1 ± 12.0
Hyperglycemia Erroneous (%)	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00

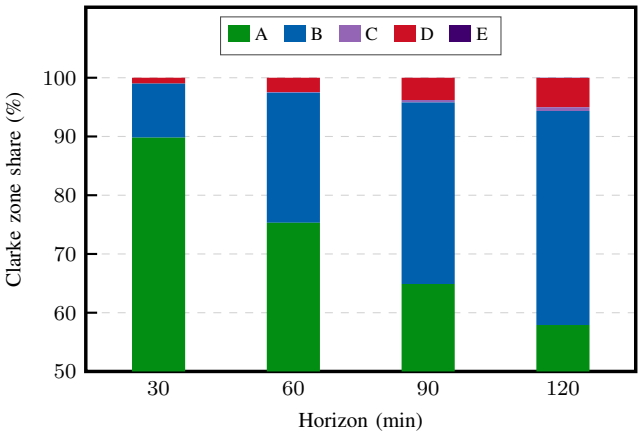


Fig. 5. *Clarke error-grid composition across 30/60/90/120-minute horizons. Zone A decreases from ∼90% to ∼58%, with the difference shifting mainly to zone B, while dangerous zones D and E remain negligible. Despite this favorable composition, the median forecast still misses many hypoglycemic events, so low D/E rates should not be interpreted as high hypoglycemia sensitivity.*

65.9% of PRED-EGA classifications were Accurate at 30 min, falling to 10.9% at 120 min (Table VIII).

G. Hypoglycemia Detection and Predictive Distribution Calibration

Point forecasts trained with squared or Huber losses regress toward the mean and underestimate hypoglycemic excursions. We therefore evaluated hypoglycemia detection in four ways: consensus event-level detection, a head-to-head comparison of A0 and A7 with quantile heads, validation-based threshold selection, and multi-horizon calibration.

1) Consensus Event-Based Hypoglycemia Evaluation: Following international consensus [58], a hypoglycemic episode was defined as at least 15 continuous minutes (three consecutive 5-minute readings) below 70 mg/dL; the test set contains 56 such episodes over 92.0 patient-days (Table IX). With 56 events and about 552 effectively independent test windows (494 using the autocorrelation time), these estimates carry substantial uncertainty. The median forecast detected 11 of 56 events (sensitivity 19.6%, specificity 99.4%, 0.62 false alarms/day, median lead time 15 min). An alarm on the lower quantile ($\hat{G}_{0.10} < 70$ mg/dL) detected 38 of 56 before onset (67.9%; specificity 95.4%, 2.21 false alarms/day, median lead time 15 min). Monitoring $\min(\hat{G}_{0.10}(30), \hat{G}_{0.10}(60)) <$ 70 mg/dL detected 46 of 56 (82.1%; specificity 92.1%, 3.46 false alarms/day) and extended the median lead time to

25.0 min.

2) Quantile Alarm for Baseline A0 vs. Physics A7: To test whether physics worsens hypoglycemia detection when both models have quantile heads, A0 was retrained with the same non-crossing quantile head and compared with A7 (Table IX). With the median forecast, A7's sensitivity was numerically higher than A0's (pointwise 0.551 vs. 0.413; event-level 19.6% vs. 10.7%) at similar specificity (99.3% vs. 99.4%). With the $q = 0.10$ alarm, A7 was again numerically more sensitive (pointwise 0.918 vs. 0.861; event-level 67.9% vs. 58.9%, i.e., 38 vs. 33 of 56 events) at comparable specificity (95.4% vs. 95.9%). A subject-level bootstrap (5,000 resamples over the 12 test subjects) gave 95% CIs for this event-level sensitivity of [0.585, 0.750] for A7 and [0.453, 0.726] for A0; the CI for the paired difference, $[-0.050, 0.235]$, includes zero, so the difference is not statistically significant. In this comparison there is therefore no evidence that physics degrades hypoglycemia detection, but no significant improvement either.

Single-run hypoglycemia estimates are also sensitive to training. In the single-seed ablation without a quantile head (Table VI), A7 had lower sensitivity than A0 (0.608 vs. 0.637) and a larger hypoglycemic bias (+8.61 vs. +7.29 mg/dL), and A0's median-forecast sensitivity fell from 0.637 there to 0.413 once a quantile head was added. Tables VI and IX therefore report different runs and architectures and should not be compared directly.

3) Validation Threshold Selection and Operating Curves: To avoid test-set reuse, the alarm threshold was chosen on the validation split by maximizing the F_2 score, which weights sensitivity twice as heavily as precision. The selected threshold, $T^* = 84.0$ mg/dL ($F_2 = 0.771$), was evaluated once on the test split. With true hypoglycemia ($G < 70$ mg/dL) as the target, alarming at $\hat{G}_{0.10} < 84$ mg/dL gave sensitivity 0.988, specificity 0.868, precision 0.175, and 3,397 false positives, compared with 0.918, 0.951, and 0.347 (1,257 false positives) for the standard 70 mg/dL alarm. If the target is also redefined as $G < 84$ mg/dL, the same alarm gives sensitivity 0.943, specificity 0.908, and precision 0.451. The higher threshold therefore trades precision for sensitivity. Only $q \in \{0.1, 0.5, 0.9\}$ are trained (Eq. (16)); the levels $q = 0.05$ and 0.20 were obtained by treating the lower half of the predictive distribution as Gaussian, with $\sigma_L = (\hat{G}_{0.5} - \hat{G}_{0.1})/(-\Phi^{-1}(0.10))$ and $\hat{G}_p = \hat{G}_{0.5} + \sigma_L \, \Phi^{-1}(p)$, an approximation rather than independently trained quantiles. Sensitivity–precision curves for $q \in \{0.05, 0.10, 0.20\}$ (`data/sensitivity_precision_curves.csv`) show that the operating point can be adjusted along a continuous trade-off.

4) Calibration and Uncertainty Quantification: Table X reports calibration, sharpness, and scoring rules. Empirical coverage of the nominal 80% interval ($[q_{0.10}, q_{0.90}]$) was 79.4%, 78.5%, 77.2%, and 76.9% at 30, 60, 90, and 120 min, and the lower quantile covered 9.47% of observations at 30 min and 9.96% at 120 min, close to the nominal 10.0%. The upper quantile, in contrast, was under-covered at every horizon: 88.9%, 88.1%, 87.4%, and 86.9% of observations fell below it at 30, 60, 90, and 120 min, against a nominal 90%. For glucose below 80 mg/dL, the interval covered 76.4% of observations and the lower quantile 22.4%. Between 30 and 120 min, mean interval width grew from 39.9 to 96.7 mg/dL and CRPS from 8.43 to 21.01 mg/dL. With only three trained quantiles, CRPS was approximated by trapezoidal integration of twice the pinball loss over $\tau \in \{0, 0.1, 0.5, 0.9, 1\}$ (zero at the ends): $0.5\,\text{pinball}_{0.1} + 0.8\,\text{pinball}_{0.5} + 0.5\,\text{pinball}_{0.9}$.

Point forecasts also increasingly compress glucose excursions at longer horizons. The predicted-to-actual coefficient-of-variation ratio decreases from 0.973 to 0.802, with TIR increasingly overestimated and TBR underestimated (Fig. 6). Consistently, predicted LBGI falls increasingly below the actual value, from 0.65 versus 0.77 at 30 minutes to 0.33 versus 0.74 at 120 minutes, indicating progressively weaker representation of hypoglycemic risk in longer-horizon point forecasts.

H. Patient-Specific Insulin Sensitivity and Validity Checks

Three pre-specified checks determine whether S_I may be interpreted physiologically (Table XI). All pass under the official protocol: between-subject variation is non-degenerate (CV $= 35.8\%$), test–retest reliability is high (ICC$(1) = 0.890$), and S_I is strongly inversely related to total daily insulin dose ($\rho = -0.839$, 95% CI $[-0.99, -0.44]$). Under LOSO, stability and variability remain, but the dose correlation fails. Hence S_I is a stable, subject-specific estimate whose dose relation holds only in the personalized setting. Because the parameter head sees the insulin inputs and total daily dose is computed from the same records, this relation is concurrent consistency, not external validity; cross-subject validity is not established.

For the stability check, each subject's window-level S_I estimates were split into four disjoint, time-ordered blocks, each reduced to its median, and ICC(1) was computed across subjects on these four block medians. Each subject's S_I is the median of its window-level estimates; the between-subject CV and the dose correlation are computed across these 12 subject medians. Total daily dose was the total recorded insulin delivery divided by the number of recorded days (U/day).

1) Addressing Potential Circularity and Confounding: We tested whether the stability and dose correlation of S_I could be artifacts of the temporal-consistency loss or of ambient glucose, and whether S_I agrees with classical minimal-model estimation (Table XII):

1) **No temporal-consistency loss ($\lambda_{tc} = 0$):** Retrained without the penalty (`abl-A7-notc`), S_I still passed all checks: ICC(1) = 0.645, CV = 22.2%, all estimates within the Ward IDDM bounds, and $\rho = -0.804$ with total daily dose ($p = 0.0016$, 95% CI $[-0.955, -0.451]$). Stability is therefore partly, but not solely, attributable to the penalty: removing it lowered ICC(1) from 0.890 to 0.645 and CV from 35.8% to 22.2%.
2) **Partial correlation controlling for mean glucose:** Controlling for test-interval mean glucose, the S_I–TDD partial correlation was $r = -0.838$ (rank, $p = 0.0013$) and $r = -0.886$ (Pearson, $p = 0.0003$); controlling for whole-study mean glucose gave $r = -0.843$ ($p =$

TABLE IX

CONSENSUS EVENT-LEVEL AND POINTWISE HYPOGLYCEMIA DETECTION AT 30 MINUTES ACROSS MODELS AND ALARM HEADS (92.0 PATIENT-DAYS, 56 EPISODES). A0 WAS RETRAINED WITH THE SAME QUANTILE HEAD AS A7 (SECTION III-G2), SO ITS VALUES DIFFER FROM THE SINGLE-SEED ABLATION RUN IN TABLE VI. SUBJECT-BOOTSTRAP 95% CIS (5,000 RESAMPLES) FOR EVENT SENSITIVITY WITH THE $q = 0.10$ ALARM: A7 [0.585, 0.750], A0 [0.453, 0.726], PAIRED DIFFERENCE [−0.050, 0.235].

Model	Alarm Head	Point Sens.	Point Spec.	Point Prec.	Event Sens.	Specificity	False Alarms/Day	Median Lead Time
A0 (No physics)	Median (point)	0.413	0.994	0.657	10.7%	99.4%	0.59	12.5 min
A0 (No physics)	Quantile $q = 0.10$	0.861	0.956	0.357	58.9%	95.9%	2.00	15.0 min
A7 (Physics hybrid)	Median (point)	0.551	0.993	0.699	19.6%	99.4%	0.62	15.0 min
A7 (Physics hybrid)	Quantile $q = 0.10$	**0.918**	0.951	0.347	**67.9%**	95.4%	2.21	15.0 min
A7 (Physics hybrid)	Multi-horizon $q = 0.10$	–	–	–	**82.1%**	92.1%	3.46	**25.0 min**

TABLE X

MULTI-HORIZON CALIBRATION, CONDITIONAL COVERAGE, INTERVAL WIDTH, AND SCORING RULES ACROSS ALL 12 SUBJECTS (26,498 TEST WINDOWS)

Horizon	Nominal 80% Cov.	Lower $q_{0.10}$ Cov.	Upper $q_{0.90}$ Cov.	Cov. ($G < 80$)	Width (mg/dL)	Pinball ($q_{0.10}$)	Pinball ($q_{0.50}$)	CRPS (mg/dL)
30 min	79.4%	9.47%	88.9%	76.4%	39.9	2.98	6.53	8.43
60 min	78.5%	9.63%	88.1%	64.1%	65.9	4.77	10.96	13.99
90 min	77.2%	10.22%	87.4%	55.7%	83.2	6.02	14.17	18.02
120 min	76.9%	9.96%	86.9%	53.3%	96.7	6.94	16.57	21.01

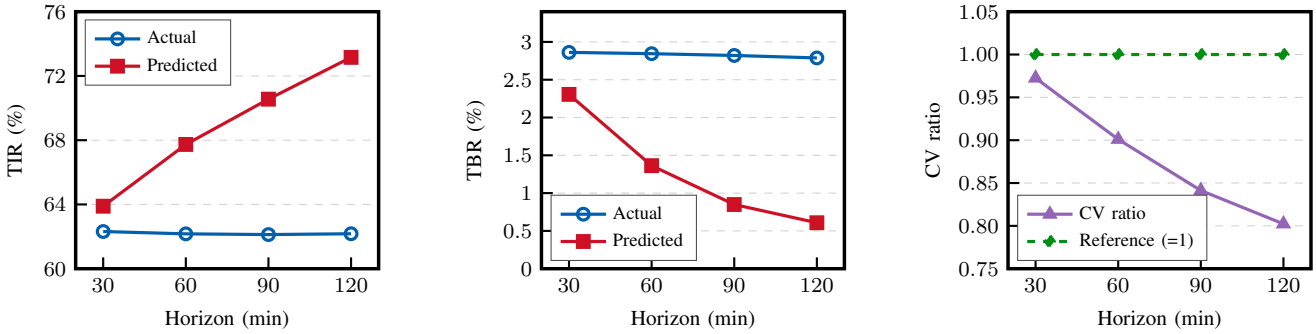


Fig. 6. *Excursion compression with horizon (30/60/90/120 minutes). a: actual time-in-range (TIR) stays flat near 62% while predicted TIR climbs toward 74%, an increasing overestimate. b: actual time-below-range (TBR) stays flat near 2.8% while predicted TBR falls toward 0.6%, an increasing underestimate of hypoglycemic exposure. c: the predicted-to-actual coefficient-of-variation ratio declines from about 0.97 to 0.80. All three panels show the same mechanism: as the horizon lengthens, the point forecast increasingly regresses toward the mean and loses the variance of the true glucose trace.*

TABLE XI

PRE-SPECIFIED CHECKS OF INSULIN SENSITIVITY

Check	Threshold	Official	LOSO
Between-subject CV	>10%	35.8%	34.3%
Test–retest ICC(1)	>0.5	0.890	0.816
ρ vs. total daily dose	<0, CI excludes 0	−0.839	−0.357
95% CI for ρ	–	[−0.99, −0.44]	[−0.82, 0.28]

0.0011) and $r = -0.875$ ($p = 0.0004$). Ambient glucose does not explain the relationship.

3) **Independent minimal-model fit:** S_I from the parameter head agreed with per-subject Bergman estimates obtained by nonlinear least squares/MAP (Spearman $\rho = 0.790$, $p = 0.0022$; Pearson $r = 0.831$, $p = 0.0008$), without iterative fitting at inference.
4) **Insulin-to-carbohydrate ratio:** For the six 2018 subjects with clinical ICR (grams of carbohydrate per unit of insulin, so a positive correlation is expected), the correlation with S_I was Spearman $\rho = +0.371$ ($p = 0.469$) and Pearson $r = +0.079$ ($p = 0.881$); with $n = 6$, this check is underpowered.
5) **Learned trust gate g:** The gate on the Bergman prior, initialized at $g_0 \approx 0.10$, converged to 0.152 in the point forecaster (A7) and 0.128 in the quantile forecaster.

Body weight is not identifiable because all OhioT1DM files use the same de-identification placeholder; a fixed nominal body weight of 70 kg is therefore used for every subject wherever weight is required (the per-kilogram volumes V_G and V_I and the glucose appearance rate R_a). The dose-per-kilogram correlation is therefore not independent of total daily dose. Moreover, p_2 and p_3 are range-constrained, so agreement with a plausible interval is partly imposed rather than learned.

I. Feature Attribution

IG and permutation importance agree with Spearman $\rho = 0.731$ ($p = 6.05 \times 10^{-7}$, $n = 35$) and share three of their top five features. Both rank five-minute glucose rate of change first

TABLE XII
NON-CIRCULARITY, CONFOUNDING, AND PHYSIOLOGICAL VALIDATION OF LEARNED INSULIN SENSITIVITY S_I

Analysis / Validation Check	Value	Interpretation / Threshold
Temporal Consistency Ablation ($\lambda_{tc} = 0.0$):		
Between-subject CV	22.2%	Non-degenerate ($> 10\%$)
Test–retest reliability ICC(1)	0.645	Stable without penalty (> 0.5)
Spearman ρ vs. TDD	-0.804	Significant ($p = 0.0016$, 95% CI $[-0.96, -0.45]$)
Ward IDDM range adherence	100%	Within bounds (partly imposed by design)
Confounding by Ambient Glucose:		
Unadjusted $\rho(S_I, \text{TDD})$	-0.839	Baseline ($p = 0.0006$)
Partial rank $r(S_I, \text{TDD} \mid \bar{G}_{\text{test}})$	-0.838	After adjustment ($p = 0.0013$)
Partial Pearson $r(S_I, \text{TDD} \mid \bar{G}_{\text{test}})$	-0.886	After adjustment ($p = 0.0003$)
Partial rank $r(S_I, \text{TDD} \mid \bar{G}_{\text{all}})$	-0.843	After adjustment ($p = 0.0011$)
Agreement with Classic Minimal Model:		
Agreement with NLS/MAP Bergman (Spearman ρ)	$+0.790$	Agreement ($p = 0.0022$)
Agreement with NLS/MAP Bergman (Pearson r)	$+0.831$	Agreement ($p = 0.0008$)
Agreement with clinical ICR ($n = 6$, exploratory)	$+0.371$	Underpowered ($p = 0.469$)
Mechanistic Prior Trust Gate:		
Initialized gate $g_0 = \sigma(-2.2)$	9.98%	Initialization
Converged gate g (Point Forecaster, A7)	15.2%	Converged value
Converged gate g (Quantile Forecaster, A7)	12.8%	Converged value

by a wide margin. Permuting it raises MAE by 3.11 mg/dL, over seven times the next feature. The group-level IG result in Fig. 7 assigns 42.7% to glucose-derived features and 27.1% to mechanistic states, showing that physiology-derived inputs are informative in their own right.

Attribution concentrates in the most recent input positions, supporting attention pooling. The IG completeness discrepancy has median relative magnitude 3.2% and p_{95} 70%; it does not shrink with more integration steps and remains unresolved. Conclusions are therefore restricted to findings corroborated by permutation importance.

J. Comparison With Published Work

At 30 minutes, the current work is mid-field relative to the 2020 challenge literature [67], [71-74] (Table XIII): on the matched 2020 cohort (subjects 540/544/552/567/584/596, the six subjects on which the challenge entries were scored), RMSE is 18.74 and MAE is 13.24 mg/dL. Further 2020 entries report RMSE@30 of 18.34 and 19.60 mg/dL [75-76]. At 60 minutes, RMSE (31.16) is essentially tied with the best entry (31.10), and MAE (22.55) is lower than every entry except one whose MAE-to-RMSE ratio is far outside the range of the other entries [72]; differences in window eligibility, missing-data handling, external pretraining, and best-configuration reporting prevent a state-of-the-art claim. Across all 12 subjects, the only published comparison is GARNN [78] (18.97/13.34 mg/dL at 30 minutes), against which the all-subject result (18.84/13.08) should be read as a tie. The defensible advances are calibrated hypoglycemia detection, explicit personalized-versus-LOSO decomposition, pre-specified checks of a physiological parameter, and a quantified account of the benefits and costs of Bergman constraints. The all-12-subject official figures reported throughout the Results section (e.g., 18.84/13.08 at 30 minutes) differ slightly from the 2020-cohort figures in Tables I and XIII; both are correct, answer different comparability questions, and should not be averaged together.

IV. DISCUSSION

The results separate three concepts that are often conflated in digital-twin and glucose-prediction literature [2]. First, a mechanistic constraint can be useful beyond point accuracy. Here it improves accuracy only modestly (0.51–2.38 mg/dL MAE across horizons over five seeds), but it also supplies interpretable state variables and a stable S_I estimate, and mechanistic features account for more than one quarter of total IG attribution. Second, a favorable error grid does not demonstrate event safety. The median forecast has excellent Clarke A+B coverage but inadequate hypoglycemia sensitivity; a calibrated predictive tail is required. Third, personalized temporal validation and cross-subject generalization answer different questions. The growing official–LOSO gap shows that access to a subject's history matters increasingly at longer horizons. Prospective digital-twin trials demonstrate the longer-term potential of individualized replay and adaptation, but do not remove the need to validate the forecasting layer independently [17].

The ablation also exposes a structural limitation. Bergman's equilibrium at G_b pulls trajectories toward the center of the glucose distribution. In the best configuration this adds approximately 1.3 mg/dL to an already-present regression-to-the-mean bias during hypoglycemia. This behavior appears whether physics enters as a penalty or a prior. A future model should modify the physiological structure rather than hide the bias with an asymmetric point loss.

V. LIMITATIONS AND FUTURE WORK

The cohort contains only 12 subjects and approximately 2,800 effectively independent observations, of which only about 550 fall in the test set; the hypoglycemia analysis rests on 56 events. The ablation screen used a single seed (42); only A0 and A7 were replicated over five seeds, post hoc (Section III-E1). OhioT1DM has no clamp measurement of S_I, so validation is indirect, and the dose relation fails under LOSO. Subject 552 has only 59.7% test CGM coverage, and

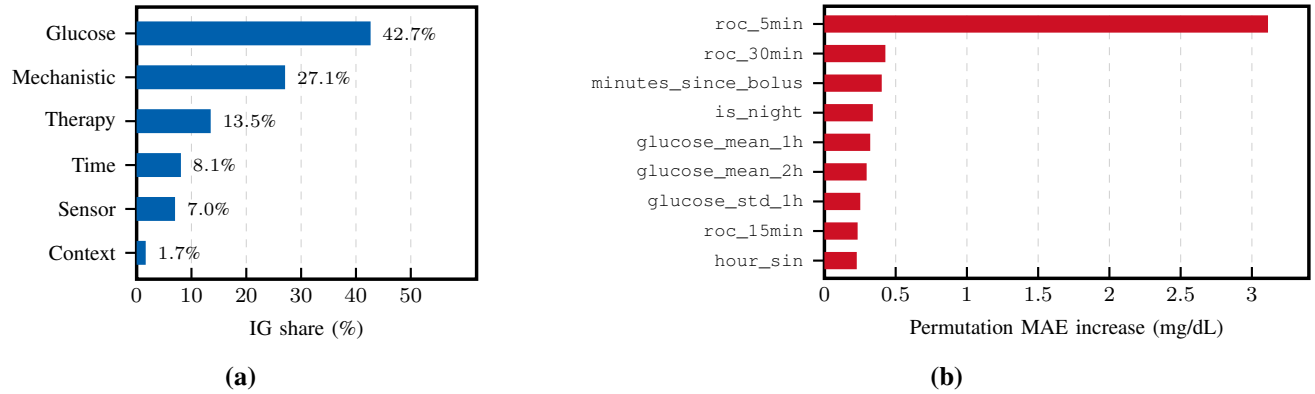


Fig. 7. *Feature attribution at 30 min using integrated gradients (a) and permutation importance (b). Glucose-derived features dominate IG attribution (42.7%), while* `roc_5` *is the strongest permuted feature (∼3.1 mg/dL MAE increase). Both methods identify glucose dynamics as the leading predictors.*

TABLE XIII
PUBLISHED OHIOT1DM COMPARISON, 2020 CHALLENGE COHORT (mg/dL)

Method	R30	A30	R60	A60	R90	A90	R120	A120
Freiburghaus CNN/LSTM [75]	17.45	11.22	33.67	23.25	–	–	–	–
Rubin-Falcone N-BEATS+BiLSTM [74]	18.22	12.83	31.66	23.60	–	–	–	–
Bevan–Coenen LSTM [73]	18.23	14.37	31.10	25.75	–	–	–	–
Pavan NN-EIM [72]	18.63	10.08	32.27	17.69	–	–	–	–
Yang MS-LSTM [67]	19.05	13.50	32.03	23.83	–	–	–	–
Current Work, official	**18.74**	**13.24**	**31.16**	**22.55**	**39.52**	**29.14**	**45.39**	**34.03**
Current Work, LOSO	20.93	14.98	34.55	25.45	43.50	32.69	49.36	37.76
Persistence	24.22	17.45	40.11	29.51	51.15	38.34	58.98	44.83

–: not reported; the Blood Glucose Level Prediction (BGLP) Challenge required only 30- and 60-minute forecasts. Current Work and Persistence rows use the same six 2020-cohort subjects as the published entries; all-12-subject results are in Table IV.

subject 567 has no carbohydrate records in the test interval. Body weight is unavailable, so a fixed 70 kg is assumed for every subject. The alarm operating point was selected on validation data without test-set reuse (Section III-G3), but lacks an elicited clinical cost ratio. No counterfactual intervention is validated, so the system is a physics-guided forecaster rather than a digital twin or treatment simulator.

The parameter head estimates a ten-component physiological vector θ_p for each subject, rather than only S_I. Table XIV lists its components: p_1, p_2, p_3, n, V_G, V_I, $t_{\max,I}$, k_{gri}, k_{abs}, and f.

Pre-specified validity checks (Table XI and Table XII) were carried out for S_I alone, for three concrete reasons rather than by omission. First, S_I is the one component with a direct clinical measurement analogue: a euglycemic clamp measures exactly this quantity, giving a real external target to validate against, whereas the remaining parameters have no equivalent bedside measurement available in OhioT1DM. Second, S_I has literature-sourced plausible bounds specific to insulin-dependent diabetes [45], giving its sigmoid-mapped range a population anchor that is not equally well established for every other compartment parameter. Third, the ablation (Table VI, A3 vs. A4) already shows that per-patient parameter estimation is accuracy-neutral overall (0.07–0.76 mg/dL), so the contribution claimed here is the stability and dose consistency of one physiological quantity, not evidence that the other nine are unimportant. One concrete obstacle already blocks part of this extension: body weight is unavailable because every OhioT1DM file carries the same de-identification placeholder, which is why a body-weight-normalized analysis was not attempted for any parameter, not only S_I.

Priority extensions are external validation on larger cohorts (e.g., Shanghai, DiaTrend), formal identifiability analysis of the remaining patient parameters, pre-specified validation of the remaining Bergman and absorption parameters (p_1, n, V_G, V_I) against literature-sourced population anchors comparable to those used for S_I, time-varying S_I, a joint multi-horizon predictive distribution, decision-theoretic alarm selection with patient-specific cost matrices, longer context, and prospective validation of counterfactual meal and insulin interventions.

VI. CONCLUSION

An integrity-checked, hybrid physics-guided forecaster can outperform a validated persistence baseline by 19–21% RMSE across 30–120-minute horizons. Physics improves point accuracy only modestly, and significantly only in a post hoc five-seed replication. The strongest evidence is instead that physiology can provide stable patient-specific structure alongside this small accuracy gain, and that calibrated distributional forecasting can recover hypoglycemic events missed by an otherwise accurate median forecast. Horizon integrity, subject-level statistics, matched generalization protocols, and explicit safety trade-offs are prerequisites for credible glucose forecasting.

TABLE XIV
PATIENT-PARAMETER VECTOR θ_p AND ADMISSIBLE INTERVALS

Symbol	Meaning	Range	Unit	Population anchor
p_1	Insulin-independent glucose disposal	0–0.030	min^{-1}	0.013
p_2	Decay of remote insulin action	0.005–0.10	min^{-1}	0.025
p_3	Build rate of action from insulin excess	10^{-6}–3×10^{-5}	mL $\mu\text{U}^{-1}\text{min}^{-2}$	6.25×10^{-6}
n	Plasma insulin clearance	0.08–0.25	min^{-1}	0.138
V_G	Glucose distribution volume	1.4–2.4	dL/kg	1.88
V_I	Insulin distribution volume	0.08–0.18	L/kg	0.12
$t_{\max,I}$	Time to peak SC absorption	30–90	min	55
k_{gri}	Gastric emptying rate	0.008–0.10	min^{-1}	0.035
k_{abs}	Intestinal absorption rate	0.005–0.10	min^{-1}	0.0167
f	Carbohydrate bioavailability	0.70–1.00	dimensionless	0.90

APPENDIX A
NOTATION, VARIABLES, AND TERMINOLOGY

This appendix summarizes the notation and feature groups used in the manuscript; full notation tables, including abbreviations, are in the Supplementary Material. Rate constants are expressed per minute unless stated otherwise. A subscript b denotes a basal or fasting quantity, a hat denotes a prediction, bold lowercase symbols denote vectors, and bold uppercase symbols denote matrices.

A. Physiological States and Inputs

Glucose G is propagated separately from a six-state linear compartment model. Insulin on board and carbohydrate on board are derived identities, not additional fitted states (cf. Eqs. (6)–(7), (9)–(10)):

$$\text{IOB} = S_1 + S_2, \qquad \text{COB} = Q_{\text{sto}} + Q_{\text{gut}}. \tag{19}$$

B. Estimated Physiological Parameters

The parameter head emits unconstrained values z_i, which are mapped to admissible, physiologically sourced intervals [45] by

$$\theta_i = \ell_i + (u_i - \ell_i)\sigma(z_i). \tag{20}$$

The bounds prevent nonphysical values but also mean that plausible range alone cannot establish identifiability.

The reported derived quantity is

$$S_I = p_3/p_2, \tag{21}$$

the steady-state gain from plasma insulin above basal to remote insulin action, already introduced as Eq. (5) [1], [44]. Higher S_I means a given insulin excess produces stronger glucose disposal. It is interpreted as a subject-specific estimate only when its cross-subject variation, test–retest ICC, and dose correlation meet the pre-specified criteria.

The implementation-level objective, following Eq. (17), can be written more explicitly as

$$\mathcal{L} = \mathcal{L}_{\text{Huber}} + \lambda_q \mathcal{L}_{\text{pinball}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} + \lambda_{\text{prior}} \mathcal{L}_{\text{prior}} + \lambda_{\text{tc}} \mathcal{L}_{\text{temporal}}. \tag{22}$$

C. Feature Contract

The 35-input feature contract contains six groups: nine glucose features (current glucose, three rates of change, and rolling summaries); nine mechanistic features (IOB, COB, the six compartment states, and glucose appearance); four therapy/timing features; three cyclic-time features; six sensor values or availability masks; and four behavioral/context indicators. Glucose features account for 42.7% of IG magnitude, mechanistic features 27.1%, therapy 13.5%, time 8.1%, sensors 7.0%, and context 1.7%. Every sensor channel has an availability indicator so missingness is not represented as a physiologically extreme standardized zero. Input interpolation is explicitly flagged, whereas forecast targets are always real CGM observations.

Table XIV lists the estimated patient-parameter vector. Notation tables for physiological states and inputs, architecture, discretization and uncertainty, training losses, evaluation metrics, and abbreviations are provided in the Supplementary Material.

REFERENCES

[1] R. N. Bergman, Y. Z. Ider, C. R. Bowden, and C. Cobelli, "Quantitative estimation of insulin sensitivity," *Am. J. Physiol.*, vol. 236, no. 6, pp. E667–E677, 1979.

[2] G. Cappon and A. Facchinetti, "Digital twins in type 1 diabetes: A systematic review," *J. Diabetes Sci. Technol.*, vol. 19, no. 1, 2025.

[3] E. Acuña and R. Aparicio, "Predicting the blood glucose level using transformers," in *Proc. 2023 Int. Conf. Computational Science and Computational Intelligence (CSCI)*, 2023, pp. 1392–1399, doi: 10.1109/CSCI62032.2023.00228.

[4] A. Marx, F. Di Stefano, H. Leutheuser, K. Chin-Cheong, M. Pfister, M.-A. Burckhardt, S. Bachmann, and J. E. Vogt, "Blood glucose forecasting from temporal and static information in children with T1D," *Front. Pediatr.*, vol. 11, Art. no. 1296904, 2023, doi: 10.3389/fped.2023.1296904.

[5] Q. Bian, A. As'arry, X. Cong, K. A. Md Rezali, and R. M. K. Raja Ahmad, "A hybrid Transformer-LSTM model applied to glucose prediction," *PLOS ONE*, vol. 19, no. 9, Art. no. e0310084, 2024.

[6] X. Xiong, X. Yang, Y. Cai, Y. Xue, J. He, and H. Su, "Exploring the potential of deep learning models integrating transformer and LSTM in predicting blood glucose levels for T1D patients," *Digit. Health*, vol. 11, Art. no. 20552076251328980, 2025, doi: 10.1177/20552076251328980.

[7] C. Duckworth, M. J. Guy, *et al.*, "Explainable machine learning for real-time hypoglycemia and hyperglycemia prediction and personalized control recommendations," *J. Diabetes Sci. Technol.*, vol. 18, no. 1, 2024.

[8] S. L. Cichosz, S. S. Olesen, and M. H. Jensen, "Explainable machine-learning models to predict weekly risk of hyperglycemia, hypoglycemia, and glycemic variability in patients with type 1 diabetes based on continuous glucose monitoring," *J. Diabetes Sci. Technol.*, vol. 20, no. 3, pp. 836–847, 2026 (published online 8 Oct. 2024), doi: 10.1177/19322968241286907.

[9] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *J. Comput. Phys.*, vol. 378, pp. 686–707, 2019.

[10] R. Pei and W. Wang, "PINN based short-term glucose prediction for type 1 diabetes," in *2025 4th Asia Conf. Algorithms, Computing and Machine Learning (CACML)*, Guangzhou, China, 2025, pp. 1–6.

[11] W. Wang, R. Pei, D. Li, S. Liu, Y. Geng, and S. Wang, "A physics-informed glucose-insulin neural network model for glucose prediction," *Tsinghua Sci. Technol.*, 2026.

[12] P. A. Mongini, M. Drecogna, and C. Toffanin, "New physics-LSTM hybrid models for control-oriented glucose prediction in Type 1 Diabetes," in *2025 American Control Conf. (ACC)*, Denver, CO, USA, 2025, pp. 662–667.

[13] P. Colmegna, K. Wang, J. Garcia-Tirado, and M. D. Breton, "Mapping data to virtual patients in type 1 diabetes," *Control Eng. Pract.*, vol. 103, Art. no. 104605, 2020, doi: 10.1016/j.conengprac.2020.104605.

[14] G. Cappon, M. Vettoretti, G. Sparacino, S. Del Favero, and A. Facchinetti, "ReplayBG: A digital twin-based methodology to identify a personalized model from type 1 diabetes data and simulate glucose concentrations to assess alternative therapies," *IEEE Trans. Biomed. Eng.*, vol. 70, no. 11, pp. 3227–3238, 2023, doi: 10.1109/TBME.2023.3286856.

[15] V. Roquemen-Echeverri, T. Kushner, P. G. Jacobs, and C. Mosquera-Lopez, "A physiologically-constrained neural network digital twin framework for replicating glucose dynamics in type 1 diabetes," *Neural Comput. Appl.*, vol. 38, p. 241, 2026.

[16] C. E. Builes-Montaño, L. Lema-Perez, A. Ramírez-Rincón, J. J. Zuleta-Tobón, J. C. Restrepo-Gutiérrez, H. D. Álvarez-Zapata, J. García-Tirado, *et al.*, "A digital twin-enhanced decision support system improves time-in-range in type 1 diabetes: a randomised clinical trial," *Sci. Rep.*, vol. 15, Art. no. 39738, 2025, doi: 10.1038/s41598-025-23165-x.

[17] B. Kovatchev, P. Colmegna, J. Pavan, J. Diaz Castañeda, M. Villa Tamayo, C. Koravi, G. Santini, C. Alix, M. Stumpf, and S. Brown, "Human-machine co-adaptation to automated insulin delivery: A randomised clinical trial using digital twin technology," *npj Digit. Med.*, vol. 8, 2025.

[18] A. Adjevi, A. M. Abdirashid, F. Aktaş, M. H. B. Ucar, and S. Solak, "Explainable reinforcement learning for glucose monitoring based on Shapley value analysis," *Comput. Methods Programs Biomed.*, vol. 278, 2026.

[19] A. Y. Ahmed, "Personalized glycemic control modeling using explainable artificial intelligence for optimizing diabetes management," *Scientia. Technology, Science and Society*, vol. 3, no. 1, 2026.

[20] M. Hwang, V. P. Rachim, J. Yoo, Y. Lee, and S.-M. Park, "Generalized multi task learning framework for glucose forecasting and hypoglycemia detection using simulation to reality," *npj Digit. Med.*, vol. 8, Art. no. 612, 2025, doi: 10.1038/s41746-025-01994-4.

[21] D. Darpit, K. Vyas, J. K. Jayagopal, A. Garcia, M. Erraguntla, and M. Lawley, "A personalized federated learning-based glucose prediction algorithm for high-risk glycemic excursion regions in type 1 diabetes," *Sci. Rep.*, vol. 15, Art. no. 38376, 2025.

[22] M. P. Barbato, G. Rigamonti, D. Marelli, and P. Napoletano, "Lightweight sequential transformers for blood glucose level prediction in type-1 diabetes," *IEEE J. Biomed. Health Inform.*, vol. 30, no. 6, pp. 5465–5475, 2026, doi: 10.1109/JBHI.2025.3633194.

[23] A. J. Rodriguez-Almeida, C. Betancort, A. M. Wägner, G. M. Callico, and H. Fabelo, "Incorporating uncertainty estimation and interpretability in personalized glucose prediction using the Temporal Fusion Transformer," *Sensors*, vol. 25, no. 15, Art. no. 4647, 2025.

[24] Y. Lu *et al.*, "A pretrained transformer model for decoding individual glucose dynamics from continuous glucose monitoring data," *Natl. Sci. Rev.*, vol. 12, no. 5, Art. no. nwaf039, 2025.

[25] S. B. Soumma and H. Ghasemzadeh, "GlyRAG: Context-aware retrieval-augmented framework for blood glucose forecasting," arXiv preprint, 2026.

[26] H. Nemat, H. Khadem, J. Elliott, and M. Benaissa, "Data fusion of activity and CGM for predicting blood glucose levels," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 120–124, 2020.

[27] H. Khadem, H. Nemat, J. Elliott, and M. Benaissa, "Multi-lag stacking for blood glucose level prediction," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 146–150, 2020.

[28] X. Sun, M. Rashid, M. Sevil, N. Hobbs, R. Brandt, M. R. Askari, A. Shahidehpour, and A. Cinar, "Prediction of blood glucose levels for people with type 1 diabetes using latent-variable-based model," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 115–119, 2020.

[29] M. Mayo and T. Koutny, "Neural multi-class classification approach to blood glucose level forecasting with prediction uncertainty visualisation," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 80–84, 2020.

[30] J. Daniels, P. Herrero, and P. Georgiou, "Personalised glucose prediction via deep multitask networks," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 110–114, 2020.

[31] N. Ma, Y. Zhao, S. Wen, T. Yang, R. Wu, R. Tao, X. Yu, and H. Li, "Online blood glucose prediction using autoregressive moving average model with residual compensation network," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 151–155, 2020.

[32] G. Cappon, L. Meneghetti, F. Prendin, J. Pavan, G. Sparacino, S. Del Favero, and A. Facchinetti, "A personalized and interpretable deep learning based approach to predict blood glucose concentration in type 1 diabetes," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 75–79, 2020.

[33] A. R. Bhimireddy, P. Sinha, B. Oluwalade, J. W. Gichoya, and S. Purkayastha, "Blood glucose level prediction as time-series modeling using sequence-to-sequence neural networks," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 125–130, 2020.

[34] A. Bertachi, L. Biagi, I. Contreras, N. Luo, and J. Vehí, "Prediction of blood glucose levels and nocturnal hypoglycemia using physiological models and artificial neural networks," in *Proc. 3rd Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2148, pp. 85–90, 2018.

[35] K. Li, C. Liu, T. Zhu, P. Herrero, and P. Georgiou, "GluNet: A deep learning framework for accurate glucose forecasting," *IEEE J. Biomed. Health Inform.*, vol. 24, no. 2, pp. 414–423, 2020, doi: 10.1109/JBHI.2019.2931842.

[36] H. Nemat, H. Khadem, J. Elliott, and M. Benaissa, "Data-driven blood glucose level prediction in type 1 diabetes: A comprehensive comparative analysis," *Sci. Rep.*, vol. 14, Art. no. 21863, 2024, doi: 10.1038/s41598-024-70277-x.

[37] M. De Bois, M. A. El Yacoubi, and M. Ammi, "GLYFE: Review

and benchmark of personalized glucose predictive models in type 1 diabetes," *Med. Biol. Eng. Comput.*, vol. 60, no. 1, pp. 1–17, 2022, doi: 10.1007/s11517-021-02437-4.

[38] K. Gu, R. Dang, and T. Prioleau, "Neural physiological model: A simple module for blood glucose prediction," in *Proc. 42nd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2020, pp. 5476–5481, doi: 10.1109/EMBC44109.2020.9176004.

[39] J. Martinsson, A. Schliep, B. Eliasson, C. Meijner, S. Persson, and O. Mogren, "Automatic blood glucose prediction with confidence using recurrent neural networks," in *Proc. 3rd Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2148, pp. 64–68, 2018.

[40] J. Xie and Q. Wang, "Benchmark machine learning approaches with classical time series approaches on the blood glucose level prediction challenge," in *Proc. 3rd Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2148, pp. 97–102, 2018.

[41] C. Marling and R. Bunescu, "The OhioT1DM dataset for blood glucose level prediction," in *Proc. 3rd Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2148, pp. 60–63, 2018.

[42] C. Marling and R. Bunescu, "The OhioT1DM dataset for blood glucose level prediction: Update 2020," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 71–74, 2020.

[43] G. V. Bayley and J. M. Hammersley, "The effective number of independent observations in an autocorrelated time series," *J. R. Stat. Soc. Suppl.*, vol. 8, no. 2, pp. 184–197, 1946.

[44] R. N. Bergman, L. S. Phillips, and C. Cobelli, "Physiologic evaluation of factors controlling glucose tolerance in man," *J. Clin. Invest.*, vol. 68, no. 6, pp. 1456–1467, 1981.

[45] G. M. Ward *et al.*, "A modified minimal model analysis of insulin sensitivity and glucose-mediated glucose disposal in insulin-dependent diabetes," *Metabolism*, vol. 40, no. 1, pp. 4–9, 1991.

[46] R. Hovorka *et al.*, "Nonlinear model predictive control of glucose concentration in subjects with type 1 diabetes," *Physiol. Meas.*, vol. 25, no. 4, pp. 905–920, 2004.

[47] E. D. Lehmann and T. Deutsch, "A physiological model of glucose–insulin interaction in type 1 diabetes mellitus," *J. Biomed. Eng.*, vol. 14, no. 3, pp. 235–242, 1992.

[48] C. Dalla Man, M. Camilleri, and C. Cobelli, "A system model of oral glucose absorption: Validation on gold standard data," *IEEE Trans. Biomed. Eng.*, vol. 53, no. 12, pp. 2472–2478, 2006.

[49] C. Dalla Man, R. A. Rizza, and C. Cobelli, "Meal simulation model of the glucose–insulin system," *IEEE Trans. Biomed. Eng.*, vol. 54, no. 10, pp. 1740–1749, 2007.

[50] A. Vaswani *et al.*, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 5998–6008.

[51] P. J. Huber, "Robust estimation of a location parameter," *Ann. Math. Statist.*, vol. 35, no. 1, pp. 73–101, 1964.

[52] R. Koenker and G. Bassett, Jr., "Regression quantiles," *Econometrica*, vol. 46, no. 1, pp. 33–50, 1978.

[53] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient flow pathologies in physics-informed neural networks," *SIAM J. Sci. Comput.*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[54] A. Kendall, Y. Gal, and R. Cipolla, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7482–7491.

[55] W. L. Clarke *et al.*, "Evaluating clinical accuracy of systems for self-monitoring of blood glucose," *Diabetes Care*, vol. 10, no. 5, pp. 622–628, 1987.

[56] J. L. Parkes, S. L. Slatin, S. Pardo, and B. H. Ginsberg, "A new consensus error grid to evaluate the clinical significance of inaccuracies in the measurement of blood glucose," *Diabetes Care*, vol. 23, no. 8, pp. 1143–1148, 2000.

[57] A. Pfützner, D. C. Klonoff, S. Pardo, and J. L. Parkes, "Technical aspects of the Parkes error grid," *J. Diabetes Sci. Technol.*, vol. 7, no. 5, pp. 1275–1281, 2013.

[58] T. Battelino *et al.*, "Clinical targets for continuous glucose monitoring data interpretation: Recommendations from the International Consensus on Time in Range," *Diabetes Care*, vol. 42, no. 8, pp. 1593–1603, 2019.

[59] B. P. Kovatchev, D. J. Cox, L. A. Gonder-Frederick, and W. L. Clarke, "Symmetrization of the blood glucose measurement scale and its applications," *Diabetes Care*, vol. 20, no. 11, pp. 1655–1658, 1997.

[60] B. P. Kovatchev *et al.*, "Evaluation of a new measure of blood glucose variability in diabetes," *Diabetes Care*, vol. 29, no. 11, pp. 2433–2438, 2006.

[61] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bull.*, vol. 1, no. 6, pp. 80–83, 1945.

[62] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Stat.*, vol. 6, no. 2, pp. 65–70, 1979.

[63] P. E. Shrout and J. L. Fleiss, "Intraclass correlations: Uses in assessing rater reliability," *Psychol. Bull.*, vol. 86, no. 2, pp. 420–428, 1979.

[64] C. Spearman, "The proof and measurement of association between two things," *Am. J. Psychol.*, vol. 15, no. 1, pp. 72–101, 1904.

[65] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proc. 34th Int. Conf. Mach. Learn.*, PMLR, vol. 70, 2017, pp. 3319–3328.

[66] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, pp. 5–32, 2001.

[67] T. Yang *et al.*, "Multi-scale long short-term memory network with multi-lag structure for blood glucose prediction," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 136–140, 2020.

[68] T. K. Koo and M. Y. Li, "A guideline of selecting and reporting intraclass correlation coefficients for reliability research," *J. Chiropr. Med.*, vol. 15, no. 2, pp. 155–163, 2016.

[69] P. Davidson, H. Hebblewhite, R. Steed, and B. Bode, "Analysis of guidelines for basal-bolus insulin dosing: Basal insulin, correction factor, and carbohydrate-to-insulin ratio," *Endocr. Pract.*, vol. 14, no. 9, pp. 1095–1101, 2008, doi: 10.4158/EP.14.9.1095.

[70] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.

[71] S. Sivananthan, V. Naumova, C. Dalla Man, A. Facchinetti, E. Renard, C. Cobelli, and S. V. Pereverzyev, "Assessment of blood glucose predictors: The prediction-error grid analysis," *Diabetes Technol. Ther.*, vol. 13, no. 8, pp. 787–796, 2011.

[72] J. Pavan *et al.*, "Personalized machine learning algorithm based on shallow network and error imputation module for an improved blood glucose prediction," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 95–99, 2020.

[73] R. Bevan and F. Coenen, "Experiments in non-personalized future blood glucose level prediction," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 100–104, 2020.

[74] H. Rubin-Falcone, I. Fox, and J. Wiens, "Deep residual time-series forecasting: Application to blood glucose prediction," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 105–109, 2020.

[75] J. Freiburghaus, A. Rizzotti-Kaddouri, and F. Albertetti, "A deep learning approach for blood glucose prediction and monitoring of type 1 diabetes patients," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 131–135, 2020.

[76] T. Zhu, X. Yao, K. Li, P. Herrero, and P. Georgiou, "Blood glucose prediction for type 1 diabetes using generative adversarial networks," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 90–94, 2020.

[77] D. Joedicke, G. Kronberger, J. M. Colmenar, S. Winkler, J. M. Velasco, S. Contador, and J. I. Hidalgo, "Analysis of the performance of genetic programming on the Blood Glucose Level Prediction Challenge 2020," in *Proc. 5th Int. Workshop Knowledge Discovery in Healthcare Data*, CEUR-WS, vol. 2675, pp. 141–145, 2020.

[78] C. Piao, T. Zhu, S. E. Baldeweg, P. Taylor, P. Georgiou, J. Sun, J. Wang, and K. Li, "GARNN: An interpretable graph attentive recurrent neural network for predicting blood glucose levels via multivariate time series," *Neural Netw.*, vol. 185, Art. no. 107229, 2025, doi: 10.1016/j.neunet.2025.107229.